

35. Ethernet Generic PTP Timer (GPTP)

35.1 Overview

Ethernet Generic PTP timer (GPTP) enables accurate synchronization of the clock in the control system.

Table 35.1 lists the GPTP specifications and Figure 35.1 shows a block diagram.

Table 35.1 GPTP specifications

Item	Function	Description
Overall function	Timer	GPTP and AVTP timer
Data provision	—	- GPTP timer Data width: 78 bits ([77:30] = second, [29:0] = nanosecond) - AVTP timer Data width: 64 bits/32 bits
Functions	Timer control	- GPTP timer incremental value control - GPTP timer offset control - Generation of GPTP and AVTP timer
	PPS	PPS signal generation
	Media clock	- Media clock capture - Media clock recovery
	Cyclic compare	Cyclic comparison to emit pulses

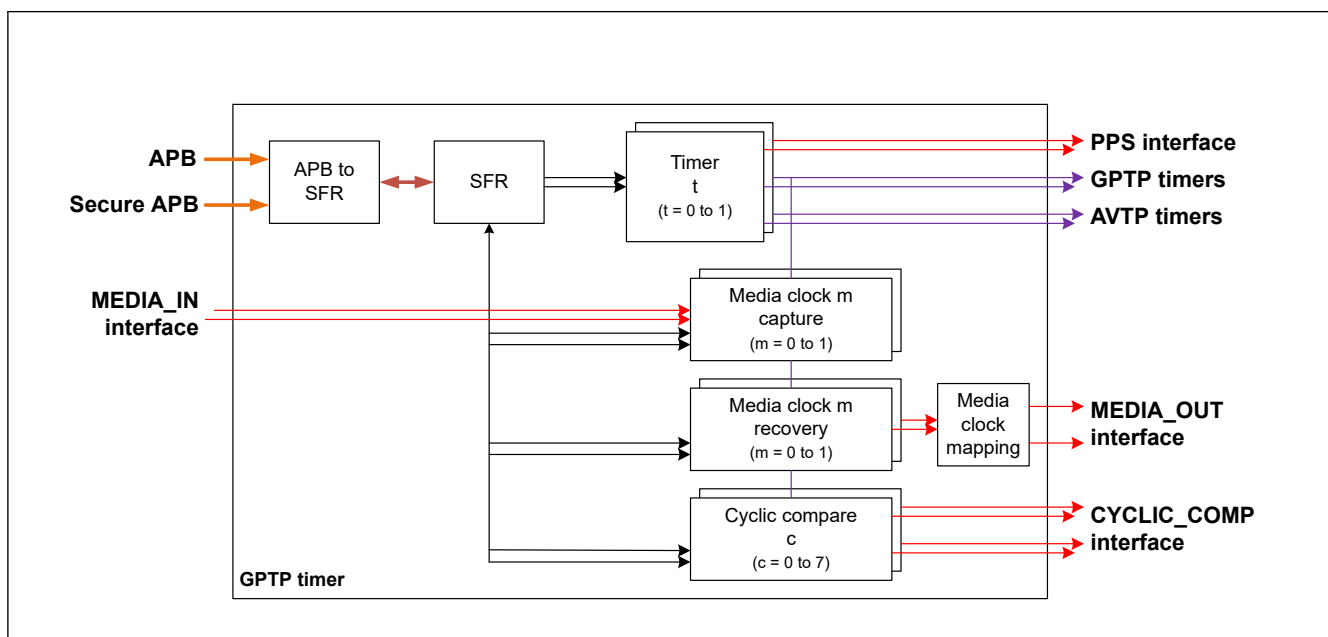


Figure 35.1 GPTP block diagram

Table 35.2 GPTP functional blocks

Block name	Function
APB to SFR	Handle SFR accesses
SFR	GPTP SFRs
Timer	Generate GPTP an AVTP timers
Media clock capture	Capture input media clocks
Media clock recovery	Recover output media clocks
Cyclic compare	Generate a cyclic pulse

35.2 gPTP Register List

Table 35.3 gPTP timer registers

Offset address	Register name	Register symbol	Write access mode
0x0000	IP Version Register	PTPIPV	—
0x0010	Timer Enable Configuration Register	PTPTMEC	—
0x0014	Timer Disable Configuration Register	PTPTMDC	—
0x0020 + 0x40 × t	Timer t Increment Value Configuration Register (t = 0, 1)	PTPTIVCt	—
0x0030 + 0x40 × t	Timer t Offset Value Configuration Register L (t = 0, 1)	PTPTOVCTl	—
0x0034 + 0x40 × t	Timer t Offset Value Configuration Register M (t = 0, 1)	PTPTOVCTm	—
0x0038 + 0x40 × t	Timer t Offset Value Configuration Register U (t = 0, 1)	PTPTOVCTu	—
0x0040 + 0x40 × t	AVTP Timer t Monitoring Register L (t = 0, 1)	PTPAVTPTMTl	—
0x0044 + 0x40 × t	AVTP Timer t Monitoring Register U (t = 0, 1)	PTPAVTPTMTu	—
0x0050 + 0x40 × t	GPTP Timer t Monitoring Register L (t = 0, 1)	PTPGTPTMTl	—
0x0054 + 0x40 × t	GPTP Timer t Monitoring Register M (t = 0, 1)	PTPGTPTMTm	—
0x0058 + 0x40 × t	GPTP Timer t Monitoring Register U (t = 0, 1)	PTPGTPTMTu	—
0x0200 + 0x10 × m	Media Clock m Capture Configuration Register (m = 0, 1)	PTPMCCCm	—
0x0204 + 0x10 × m	Media Clock m Capture Monitoring Register L (m = 0, 1)	PTPMCCMmL	—
0x0208 + 0x10 × m	Media Clock m Capture Monitoring Register M (m = 0, 1)	PTPMCCMmM	—
0x020C + 0x10 × m	Media Clock m Capture Monitoring Register U (m = 0, 1)	PTPMCCMmU	—
0x0300 + 0x10 × m	Media Clock m Recovery Configuration Register (m = 0, 1)	PTPMCRCm	—
0x0304 + 0x10 × m	Media Clock m Recovery Time Configuration Register L (m = 0, 1)	PTPMCRTCmL	—
0x0308 + 0x10 × m	Media Clock m Recovery Time Configuration Register M (m = 0, 1)	PTPMCRTCmM	—
0x030C + 0x10 × m	Media Clock m Recovery Time Configuration Register U (m = 0, 1)	PTPMCRTCmU	—
0x0400 + 0x04 × m	Media Clock m Pin Configuration Register (m = 0, 1)	PTPMPCm	—
0x0500 + 0x08 × c	Cyclic Compare c Configuration Register 0 (c = 0 to 7)	PTPCCCc0	—
0x0504 + 0x08 × c	Cyclic Compare c Configuration Register 1 (c = 0 to 7)	PTPCCCc1	—
0x0700	Interrupt Status Register 0	PTPIS0	—
0x0704	Interrupt Enable Register 0	PTPIE0	—
0x0708	Interrupt Disable Register 0	PTPID0	—
0x0710	Interrupt Status Register 1	PTPIS1	—
0x0714	Interrupt Enable Register 1	PTPIE1	—
0x0718	Interrupt Disable Register 1	PTPID1	—
0x0780	Security Configuration Register 0	PTPSCR0	—
0x0784	Security Configuration Register 1	PTPSCR1	—
0x0788	Security Configuration Register 2	PTPSCR2	—

Note: Base address: GPTP = 0x403E_0000, GPTP_NS = 0x503E_0000

35.3 gPTP Register Descriptions

35.3.1 Version Register

35.3.1.1 PTPIPV : IP Version Register

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0000

Bit position: 31

0

Bit field:

IPV[31:0]

Value after reset: 0 1 1

Bit	Symbol	Function	R/W
31:0	IPV[31:0]	IP Version	R

35.3.2 Timer Configuration Registers

35.3.2.1 PTPTMEC : Timer Enable Configuration Register

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0010

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TE1 to TE0
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	------------

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	TE1 to TE0	Timer q Enable (q = 0, 1) Bit q enables timer q. 0: Timer q disabled 1: Timer q enabled	R/W ^{*1}
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

TEq bits (Timer q Enable (q = 0, 1))

When timer q is disabled, all counters are fixed to 0.

[Setting condition]

- SW: Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to PTPTMDC.TDq clears this register bit q.

35.3.2.2 PTPTMDC : Timer Disable Configuration Register

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TD1 to TD0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	TD1 to TD0	Timer n Disable (n = 0, 1) 0: No effect 1: Bit q of this register clears PTPTMEC.TEq	W ^{*1}
31:2	—	These bits are read as 0.	R

Note 1. Read value differs from written value.

TDn bits (Timer n Disable (n = 0, 1))

[Clearing condition]

- SW: Writing 1 to bit q of this register clears PTPTMEC.TEq.

35.3.2.3 PTPTIVCt : Timer t Increment Value Configuration Register (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0020 + 0x40 × t

Bit position:	31	0																																
Bit field:	TIV[31:0]																																	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	TIV[31:0]	Timer Increment Value	R/W

TIV[31:0] bits (Timer Increment Value)

Timer increment value in ns per clock clk. PTPTIVCt.TIV[31:27] is the nanosecond part and PTPTIVCt.TIV[26:0] is the nanosecond sub-part.

[Examples]

- If clk clock frequency is 50 MHz, 100 MHz, 200 MHz or 250 MHz, PTPTIVCt.TIV should be respectively set to a value around 0xA000_0000, 0x5000_0000, 0x2800_0000 or 0x2000_0000.

35.3.2.4 PTPTOVcL : Timer t Offset Value Configuration Register L (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0030 + 0x40 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	TOVL[29:0]													
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TOVL[29:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
29:0	TOVL[29:0]	Timer Offset Value Lower Part*1	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: Only value from 0x0000_0000 to 0x3b9a_c9ff can be set to this register.

TOVL[29:0] bits (Timer Offset Value Lower Part)

- Timer offset value in GPTP format. PTPTOVcL.TOVL represents its nanosecond part.
- Writing to this register loads the new offset to timer t.
- {PTPTOVcU.TOVU, PTPTOVcM.TOVM, 0, PTPTOVcL.TOVL} is timer t offset.

35.3.2.5 PTPTOVcM : Timer t Offset Value Configuration Register M (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0034 + 0x40 × t

Bit position:	31																																	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
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Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	TOVM[31:0]	Timer Offset Value Middle Part	R/W*1

Note 1. Read value differs from written value.

TOVM[31:0] bits (Timer Offset Value Middle Part)

- Timer offset value in GPTP format. PTPTOVcM.TOVM represents its second lower 32-bit part.
- {PTPTOVcU.TOVU, PTPTOVcM.TOVM, 0, PTPTOVcL.TOVL} is timer t offset.

[Update condition]

- SW: Writing to the PTPTOVcL register updates this register to the latest value

35.3.2.6 PTPTOVctU : Timer t Offset Value Configuration Register U (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0038 + 0x40 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	TOVU[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	TOVU[15:0]	Timer Offset Value Upper Part	R/W ^{*1}
31:16	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

TOVU[15:0] bits (Timer Offset Value Upper Part)

- Timer offset value in GPTP format. PTPTOVctU.TOVU represents its second higher 16-bit part.
- {PTPTOVctU.TOVU, PTPTOVctM.TOVm, 0, PTPTOVctL.TOVl} is timer t offset.

[Update condition]

- SW: Writing to PTPTOVctL register updates this register to its latest value.

35.3.3 Timer Monitoring Registers

35.3.3.1 PTPAVTPTMtL : AVTP Timer t Monitoring Register L (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0040 + 0x40 × t

Bit position:	31																																	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																										
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Bit	Symbol	Function	R/W
31:0	AVTPL[31:0]	AVTP Timer Value Lower Part	R

AVTPL[31:0] bits (AVTP Timer Value Lower Part)

- AVTP time is in nanoseconds and derives from GPTP timer t. PTPAVTPTMtL.AVTPL represents either timer t 32-bit AVTP timer or timer t 64-bit AVTP timer lower bits.
- This register is always equal to timer t 32-bit AVTP timer or timer t 64-bit AVTP timer lower bits.

35.3.3.2 PTPAVTPTMtU : AVTP Timer t Monitoring Register U (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0044 + 0x40 × t

Bit position: 31

0

Bit field:

AVTPU[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	AVTPU[31:0]	AVTP Timer Value Upper Part	R

AVTPU[31:0] bits (AVTP Timer Value Upper Part)

AVTP time is in nanoseconds and derives from GPTP timer t. PTPAVTPTMtU.AVTPU represents timer t 64-bit AVTP timer higher bits.

[Update condition]

- HW: Reading PTPAVTPTMtL register updates this register to timer t 64-bit AVTP timer higher bits.

35.3.3.3 PTPGPTPTMtL : GPTP Timer t Monitoring Register L (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0050 + 0x40 × t

Bit position: 31

30

29

28

27

26

25

24

23

22

21

20

19

18

17

16

Bit field:

—

—

GPTPL[29:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit position: 15

14

13

12

11

10

9

8

7

6

5

4

3

2

1

0

Bit field:

GPTPL[29:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
29:0	GPTPL[29:0]	GPTP Timer Value Lower Part - PTPGPTPTMtL.GPTPL represents timer t GPTP timer nanosecond part - This register is always equal to timer t GPTP timer nanosecond part.	R
31:30	—	These bits are read as 0.	R

35.3.3.4 PTPGPTPTMtM : GPTP Timer t Monitoring Register M (t = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0054 + 0x40 × t

Bit position: 31

0

Bit field:

GTPM[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	GTPM[31:0]	GTP Timer Value Middle Part	R

GTPM[31:0] bits (GTP Timer Value Middle Part)

PTPGPTPTMtM.GTPM represents timer t GTP timer second lower 32 bit part.

[Update condition]

- HW: Reading PTPGPTPTMtL register updates this register to timer t GTP timer second lower 32-bit part.

35.3.3.5 PTPGPTPTMtU : GTP Timer t Monitoring Register U (t = 0, 1)

Base address: GTP = 0x403E_0000
GTP_NS = 0x503E_0000

Offset address: 0x0058 + 0x40 × t

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	GTPU[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	GTPU[15:0]	GTP Timer Value Upper Part	R
31:16	—	These bits are read as 0.	R

GTPU[15:0] bits (GTP Timer Value Upper Part)

PTPGPTPTMtU.GTPU represents timer t GTP timer second higher 16-bit part.

[Update condition]

- HW: Reading PTPGPTPTMtL register updates this register to timer t GTP timer second higher 16-bit part.

35.3.4 Media Clock Capture Registers**35.3.4.1 PTPMCCCm : Media Clock m Capture Configuration Register (m = 0, 1)**

Base address: GTP = 0x403E_0000
GTP_NS = 0x503E_0000

Offset address: 0x0200 + 0x10 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	MCTN S	MCTT S	MCNE E	MCPE E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MCPEE	Media Clock Capture Positive Edge Enable 0: Timer value is not captured on positive edge of MEDIA_IN[m] signal 1: Timer value is captured on positive edge of MEDIA_IN[m] signal	R/W

Bit	Symbol	Function	R/W
1	MCNEE	Media Clock Capture Negative Edge Enable 0: Timer value is not captured on negative edge of MEDIA_IN[m] signal 1: Timer value is captured on negative edge of MEDIA_IN[m] signal	R/W
2	MCTTS	Media Clock Capture Timer Type Select 0: Timer value is captured on GPTP timer 1: Timer value is captured on AVTP timer	R/W
3	MCTNS	Media Clock Capture Timer Number Select Selects from which timer the timer values are taken.	R/W
15:4	—	These bits are read as 0. The write value should be 0.	R/W
16	MCCR	Media Clock Capture Request Writing 1 to this register triggers timer capture.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

35.3.4.2 PTPMCCMmL : Media Clock m Capture Monitoring Register L (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0204 + 0x10 × m

Bit position: 31

0

Bit field:

MCCTVL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MCCTVL[31:0]	Media Clock Captured Timer Value Lower Part	R

MCCTVL[31:0] bits (Media Clock Captured Timer Value Lower Part)

- If a GPTP timer is captured, PTPMCCMmL.MCCTVL[29:0] represent the GPTP timer nanosecond part
- If an AVTP timer is captured, this register represents either the 32-bit AVTP timer or the 64-bit AVTP lower 32 bits.

[Update condition]

- Reading PTPMCCMmL register updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the next captures value
- Capturing a timer while the capture buffer is empty updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the captured value.

35.3.4.3 PTPMCCMmM : Media Clock m Capture Monitoring Register M (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0208 + 0x10 × m

Bit position: 31

0

Bit field:

MCCTVM[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MCCTVM[31:0]	Media Clock Captured Timer Value Middle Part	R

MCCTVM[31:0] bits (Media Clock Captured Timer Value Middle Part)

- If a GPTP timer is captured, this register represents the GPTP timer second part 32 lower bits

- If an AVTP timer is captured, this register represents the 64-bit AVTP higher 32 bits.

[Update condition]

- Reading PTPMCCmL register updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the next captured value
- Capturing a timer while the capture buffer is empty updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the captured value.

35.3.4.4 PTPMCCmU : Media Clock m Capture Monitoring Register U (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x020C + 0x10 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	MCCN[1:0]	—	—	—	—	—	—	MCSWC	MCNEC	MCPEC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MCCTVU[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MCCTVU[15:0]	Media Clock Captured Timer Value Upper Part	R
16	MCPEC	Media Clock Positive Edge Captured	R
17	MCNEC	Media Clock Negative Edge Captured	R
18	MCSWC	Media Clock Software Captured	R
23:19	—	These bits are read as 0.	R
25:24	MCCN[1:0]	Media Clock Capture Number	R
31:26	—	These bits are read as 0.	R

MCCTVU[15:0] bits (Media Clock Captured Timer Value Upper Part)

- If a GPTP timer is captured, this register represents the GPTP timer second part higher 16 bits
- If an AVTP timer is captured, this register is unused.

[Update condition]

- Reading PTPMCCmL register updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the next captured value
- Capturing a timer while the capture buffer is empty updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the captured value.

MCPEC bit (Media Clock Positive Edge Captured)

[Update condition]

- Reading PTPMCCmL register updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the next captured value
- Capturing a timer while the capture buffer is empty updates PTPMCCmL, PTPMCCmM and PTPMCCmU to the captured value.

MCNEC bit (Media Clock Negative Edge Captured)

[Update condition]

- Reading PTPMCCMmL register updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the next captured value
- Capturing a timer while the capture buffer is empty updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the captured value.

MCSWC bit (Media Clock Software Captured)

[Update condition]

- Reading PTPMCCMmL register updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the next captured value
- Capturing a timer while the capture buffer is empty updates PTPMCCMmL, PTPMCCMmM and PTPMCCMmU to the captured value.

MCCN[1:0] bits (Media Clock Capture Number)

[Increment condition]

- HW: Incremented by one when Media Capture Unit m captures a timer other than 2.

[Decrement condition]

- SW: Decrement by one when reading by software is executed other than 0.

35.3.5 Media Clock Recovery Registers**35.3.5.1 PTPMCRCm : Media Clock m Recovery Configuration Register (m = 0, 1)**

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0300 + 0x10 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRPL[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	MRTN S	MRAM S	MRTT S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	MRTTS	Media Clock Recovery Timer Type Select* ¹ 0: Timer value is compared to GPTP timer 1: Timer value is compared to AVTP timer	R/W
1	MRAMS	Media Clock Recovery AVTP Mode Select* ² 0: 64-bit AVTP is used for comparison 1: 32-bit AVTP is used for comparison	R/W
2	MRTNS	Media Clock Recovery Timer Number Select Selects from which timer the timer values are taken.	R/W
15:3	—	These bits are read as 0. The write value should be 0.	R/W
31:16	MRPL[15:0]	Media Clock Recovery Pulse Length Pulse length for clock recovery in clk clocks - 1.	R/W

Note 1. Restrictions:

- HW: For GPTP timer, overflow is not supported. If a time smaller than the current GPTP timer is set, the comparison instantaneously matches.

Note 2. Restrictions:

- HW: For 64-bit AVTP timer, overflow is not supported. If a time smaller than the current AVTP timer is set, the comparison instantaneously matches.

35.3.5.2 PTPMCRTCmL : Media Clock m Recovery Time Configuration Register L (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0304 + 0x10 × m

Bit position: 31

0

Bit field:

MRTL[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MRTL[31:0]	Media Clock Recovery Timer Value Lower Part	R/W

MRTL[31:0] bits (Media Clock Recovery Timer Value Lower Part)

- If the compared timer is a GPTP timer, this register represents the GPTP timer nanosecond part
- If the compared timer is a 32-bit AVTP timer, this register represents the 32-bit AVTP timer
- If the compared timer is a 64-bit AVTP timer, this register represents the 64-bit AVTP lower 32 bits
- Writing to this register loads MCCTVU, MCCTVM and MCCTVL registers for comparison in the recovery buffer.

35.3.5.3 PTPMCRTCmM : Media Clock m Recovery Time Configuration Register M (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0308 + 0x10 × m

Bit position: 31

0

Bit field:

MRTVM[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	MRTVM[31:0]	Media Clock Recovery Timer Value Middle Part	R/W

MRTVM[31:0] bits (Media Clock Recovery Timer Value Middle Part)

- If the compared timer is a GPTP timer, this register represents the GPTP timer second part 32 lower bits
- If the compared timer is a 32-bit AVTP timer, this register represents nothing relevant
- If the compared timer is a 64-bit AVTP timer, this register represents the 64-bit AVTP higher 32 bits.

35.3.5.4 PTPMCRTCmU : Media Clock m Recovery Time Configuration Register U (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x030C + 0x10 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MRBC R	—	—	—	—	—	—	—	—	—	—	MCRN[2:0]			MRTT[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MRTVU[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	MRTVU[15:0]	Media Clock Recovery Timer Value Upper Part	R/W
17:16	MRTT[1:0]	Media Clock Recovery Trigger Type 0 0: When timer match, time recovery pin is set to 1 0 1: When timer match, time recovery pin is set to 0 1 0: When timer match, time recovery pin is inverted 1 1: When timer match, a positive pulse of length PTPMCRCm.MRPL is output on time recovery pin	R/W
20:18	MCRN[2:0]	Media Clock Recovery Number Displays the number on entry in the recovery buffer.	R
30:21	—	These bits are read as 0. The write value should be 0.	R/W
31	MRBCR	Media Clock Recovery Buffer Clear Request Writing one to this register clears the internal timer buffer along with PTPMCRTCmU.MCRN register.	R

MRTVU[15:0] bits (Media Clock Recovery Timer Value Upper Part)

- If the compared timer is a GPTP timer, this register represents the GPTP timer second part 16 higher bits
- If the compared timer is a 32-bit AVTP timer, this register represents nothing relevant
- If the compared timer is a 64-bit AVTP timer, this register represents nothing relevant.

35.3.5.5 PTPMCPCm : Media Clock m Pin Configuration Register (m = 0, 1)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0400 + 0x04 × m

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MRS	PE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PE	Pin Enable*1 0: MEDIA_OUT[p] fixed to 0 1: MEDIA_OUT[p] outputs the comparison result of PTPMCPCm.MRS Recovery logic	R/W
1	MRS	Media Clock Recovery Select*1 Select Recovery logic for MEDIA_OUT[p].	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Restrictions:

- Do not change this setting on any timer enabled (PTPTMEC.TEq = 1).

35.3.6 Cyclic Compare Registers

35.3.6.1 PTPCCc0 : Cyclic Compare c Configuration Register 0 (c = 0 to 7)

Base address: GTP = 0x403E_0000
GTP_NS = 0x503E_0000

Offset address: 0x0500 + 0x8 × c

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	CCOP S	—	—	—	CCTN S
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	CCTNS	Cyclic Compare Timer Number Select Selects from which timer the cyclic compare is based on.	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
4	CCOPS	Cyclic Compare Output Pin Select CYCLIC_COMP[PTPCCc0.CCOPS][c] is selected as output for cyclic compare c.	R/W
31:5	—	These bits are read as 0. The write value should be 0.	R/W

35.3.6.2 PTPCCc1 : Cyclic Compare c Configuration Register 1 (c = 0 to 7)

Base address: GTP = 0x403E_0000
GTP_NS = 0x503E_0000

Offset address: 0x0504 + 0x8 × c

Bit position:	31	0																															
Bit field:	CCV[31:0]																																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	CCV[31:0]	Cycle Compare Value - Setting this bit to a value smaller than 32 disables the compare function - Setting this bit to a value larger or equal to 32 enables the compare function.	R/W

35.3.7 Interrupt Registers

35.3.7.1 PTPIS0 : Interrupt Status Register 0

Base address: GTP = 0x403E_0000
GTP_NS = 0x503E_0000

Offset address: 0x0700

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCOES1 to MCCOES0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCS1 to MCCS0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCCS1 to MCCS0	Media Clock n Capture Status Flag (n = 0, 1)	R/W ^{*1}
15:2	—	These bits are read as 0. The write value should be 0.	R/W
17:16	MCCOES1 to MCCOES0	Media Clock n Capture Overflow Error Status Flag (n = 0, 1)	R/W ^{*1}
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

MCCSn bits (Media Clock n Capture Status Flag (n = 0, 1))

[Setting condition]

- HW: Media Capture Unit m captured a timer.

[Clearing condition]

- SW: Writing 1 to one of these bits.

MCCOESn bits (Media Clock n Capture Overflow Error Status Flag (n = 0, 1))

[Setting condition]

- HW: Bit m is set when Media Capture Unit m captured a timer, but it cannot be stored because the timer FIFO is full (PTPMCCMmU.MCCN equal to 2).

[Clearing condition]

- SW: Writing 1 to one of these bits.

[Error recovery]

- HW: The captured timer is lost, and the next capture continues normally
- SW: The timer reading is too slow, and software should be reviewed.

35.3.7.2 PTPIE0 : Interrupt Enable Register 0

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0704

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCOEE1 to MCCOEE0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCE1 to MCCE0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCCE1 to MCCE0	Media Clock m Capture Enable (m = 0, 1) Bit m enables media clock m capture 0: Interrupt m disabled 1: Interrupt m enabled	R/W ^{*1}
15:2	—	These bits are read as 0. The write value should be 0.	R/W
17:16	MCCOEE1 to MCCOEE0	Media Clock m Capture Overflow Error Enable (m = 0, 1) Bit m enables media clock m capture overflow error. 0: Interrupt m disabled 1: Interrupt m enabled	R/W ^{*1}
31:18	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

MCCEm bits (Media Clock m Capture Enable (m = 0, 1))

[Setting condition]

- SW: Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to bit m of PTPID0.MCCD register clears the corresponding bit of this register.

MCCOEE m bits (Media Clock m Capture Overflow Error Enable (m = 0, 1))

[Setting condition]

- SW: Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to bit m of PTPID0.MCCOED register clears the corresponding bit of this register.

35.3.7.3 PTPID0 : Interrupt Disable Register 0

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0708

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCOED1 to MCCOED0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCCD1 to MCCD0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCCD1 to MCCD0	Media Clock m Capture Disable (m = 0, 1)	R
15:2	—	These bits are read as 0.	R
17:16	MCCOED1 to MCCOED0	Media Clock m Capture Overflow Error Disable (m = 0, 1)	R
31:18	—	These bits are read as 0.	R

MCCDm bits (Media Clock m Capture Disable (m = 0, 1))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding bit of PTPIE0.MCCE register.

MCCOEDm bits (Media Clock m Capture Overflow Error Disable (m = 0, 1))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding bit of PTPIE0.MCCOEE register.

35.3.7.4 PTPIS1 : Interrupt Status Register 1

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0710

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCRMS1 to MCRMS0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCRMS1 to MCRMS0	Media Clock m Recovery Match Status Flag (m = 0, 1)	R/W ^{*1}
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

MCRMSm bits (Media Clock m Recovery Match Status Flag (m = 0, 1))

[Setting condition]

- HW: The oldest timer in Media Capture Unit m is matched.

[Clearing condition]

- SW: Writing 1 to one of these bits.

35.3.7.5 PTPIE1 : Interrupt Enable Register 1

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0714

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCRME1 to MCRME0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCRME1 to MCRME0	Media Clock m Recovery Match Enable (m = 0, 1) Bit m enables media clock m recovery match. 0: Interrupt m disabled 1: Interrupt m enabled	R/W ^{*1}
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

MCRME_m bits (Media Clock m Recovery Match Enable (m = 0, 1))

[Setting condition]

- SW: Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to bit m of PTPID1.MCRMD register clears the corresponding bit of this register.

35.3.7.6 PTPID1 : Interrupt Disable Register 1

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x0718

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	MCRMD1 to MCRMD0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	MCRMD1 to MCRMD0	Media Clock m Recovery Match Disable (m = 0, 1)	R
31:2	—	These bits are read as 0.	R

MCRMDm bits (Media Clock m Recovery Match Disable (m = 0, 1))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding bit of PTPIE0.MCRME register.

35.4 Register Utilization**35.4.1 Software Flows**

SW: Follow the flows described in this section.

35.4.1.1 Software Flow Legend

Figure 35.2 describes the software flow legend.

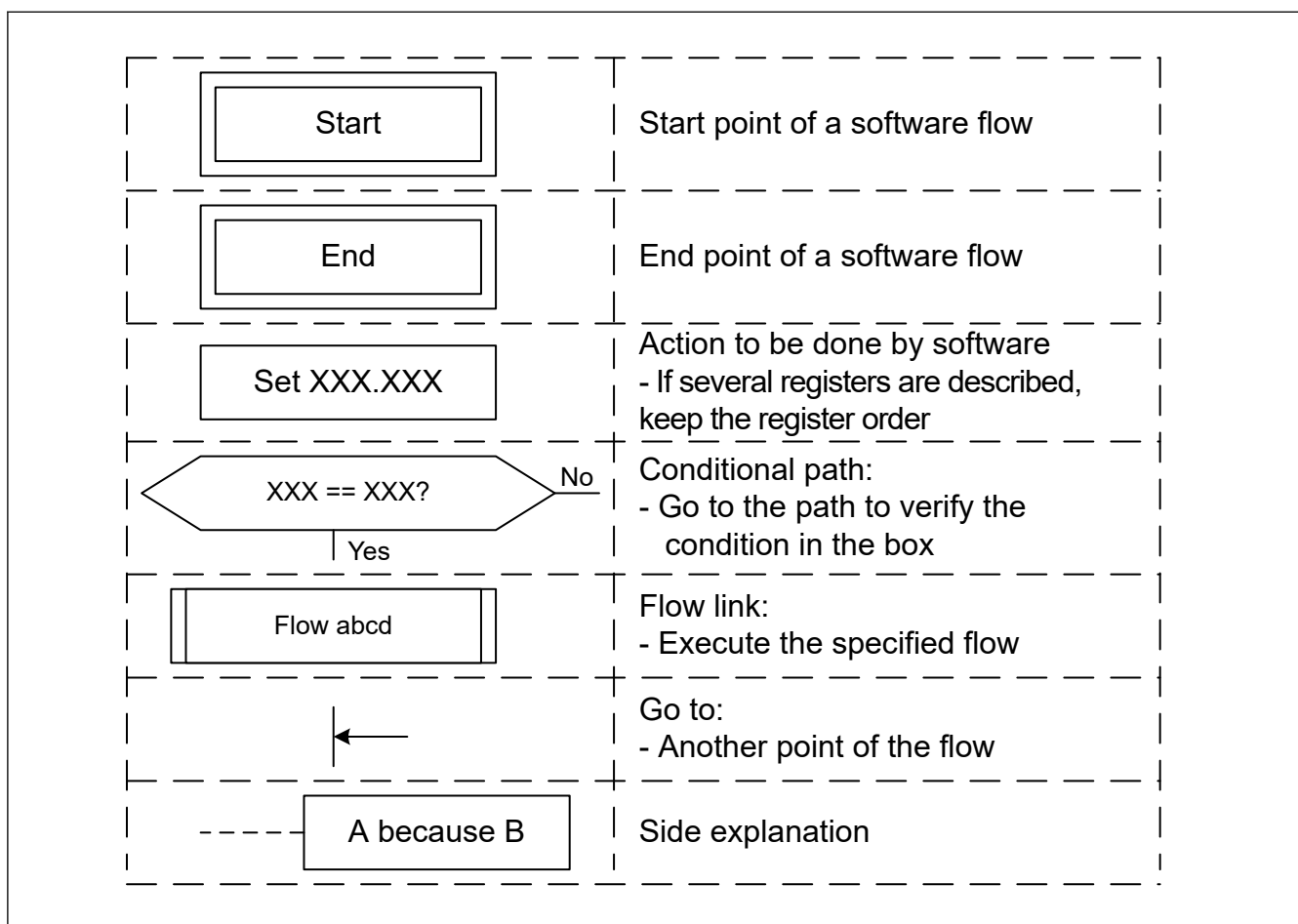
**Figure 35.2 Software flow legend****35.4.1.2 Interrupt Handling Flow**

Figure 35.3 describes the interrupt handling flow.

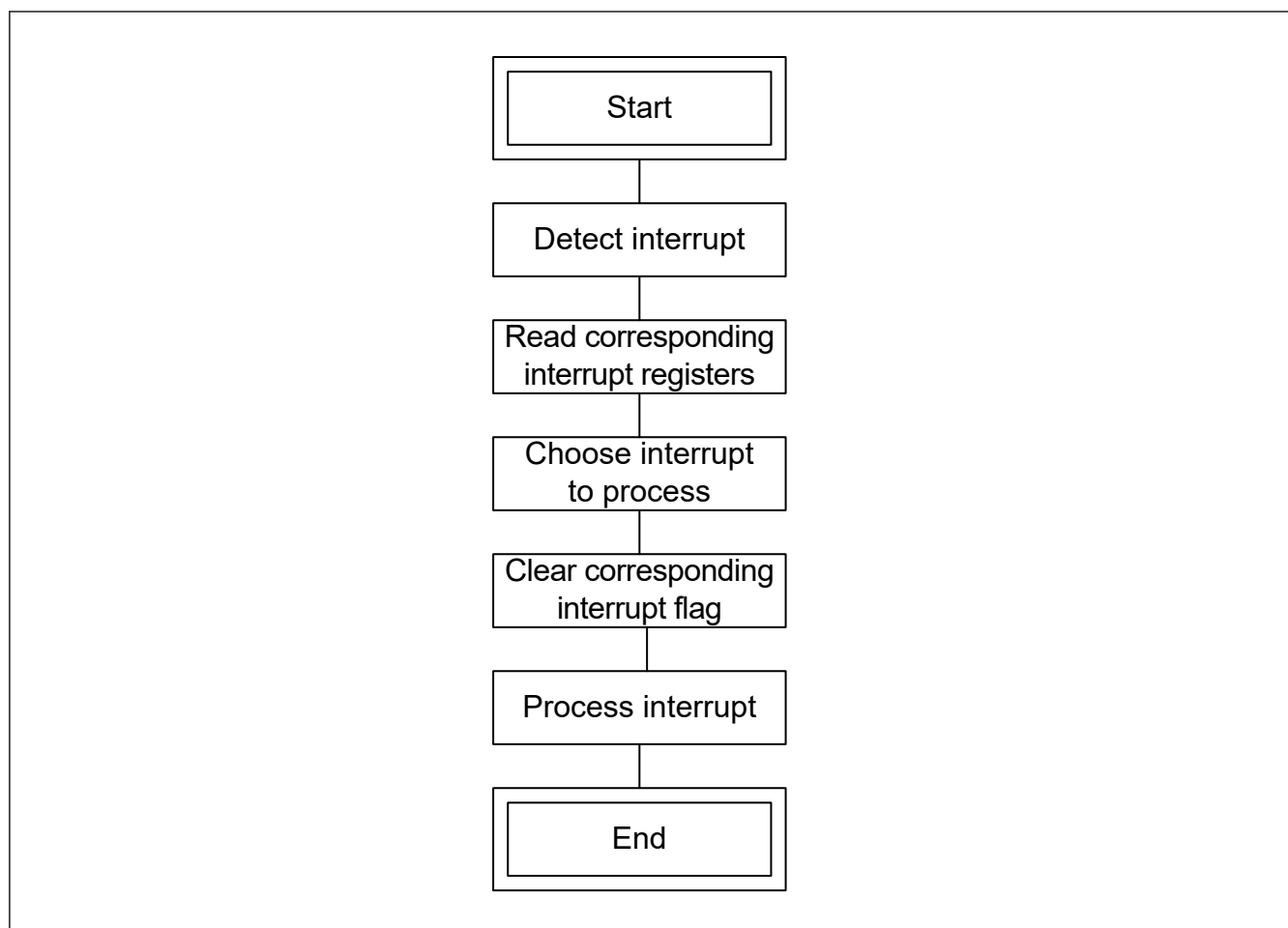


Figure 35.3 Interrupt handling flow

35.4.1.3 Timer t Flows (t = 0, 1)

This section describes the flows to set and monitor timer t.

35.4.1.3.1 Timer t Offset Setting Flow

[Figure 35.4](#) describes the timer t offset setting flow.

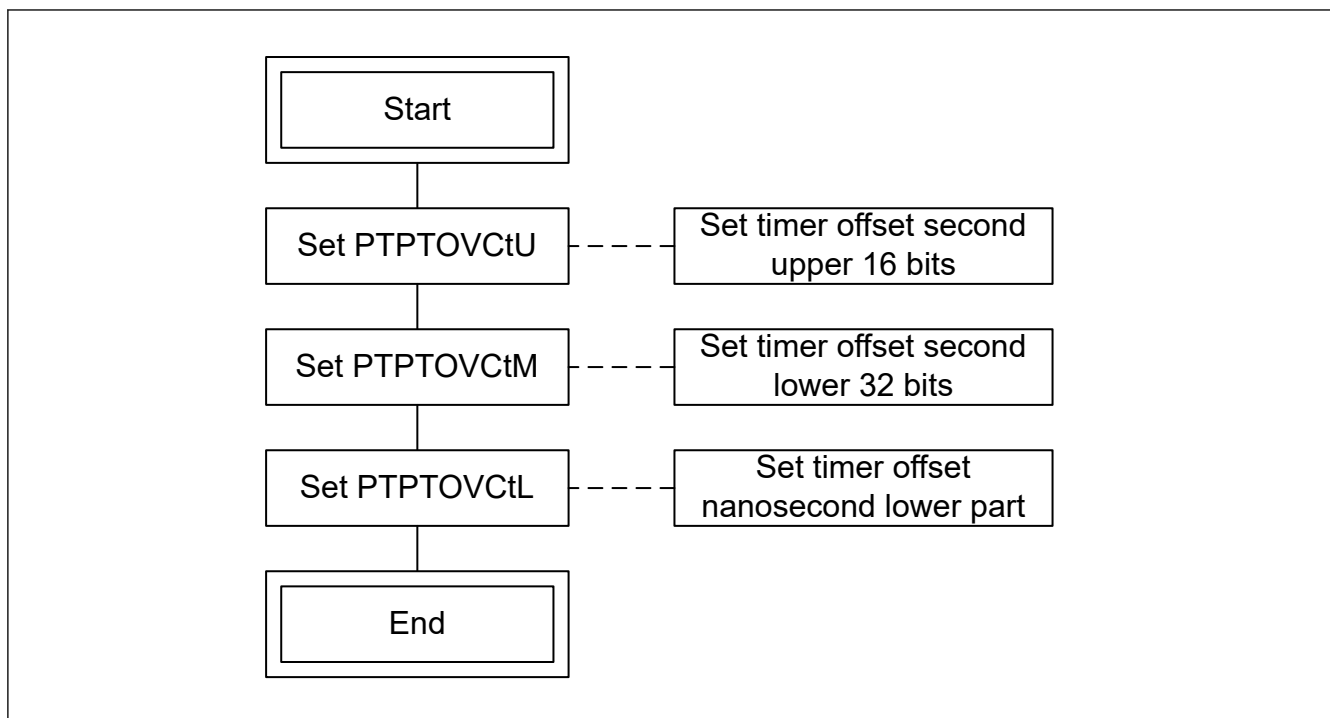


Figure 35.4 Timer t offset setting flow

35.4.1.3.2 32-bit AVTP Timer t Reading Flow

Figure 35.5 describes the 32-bit AVTP timer t reading flow.

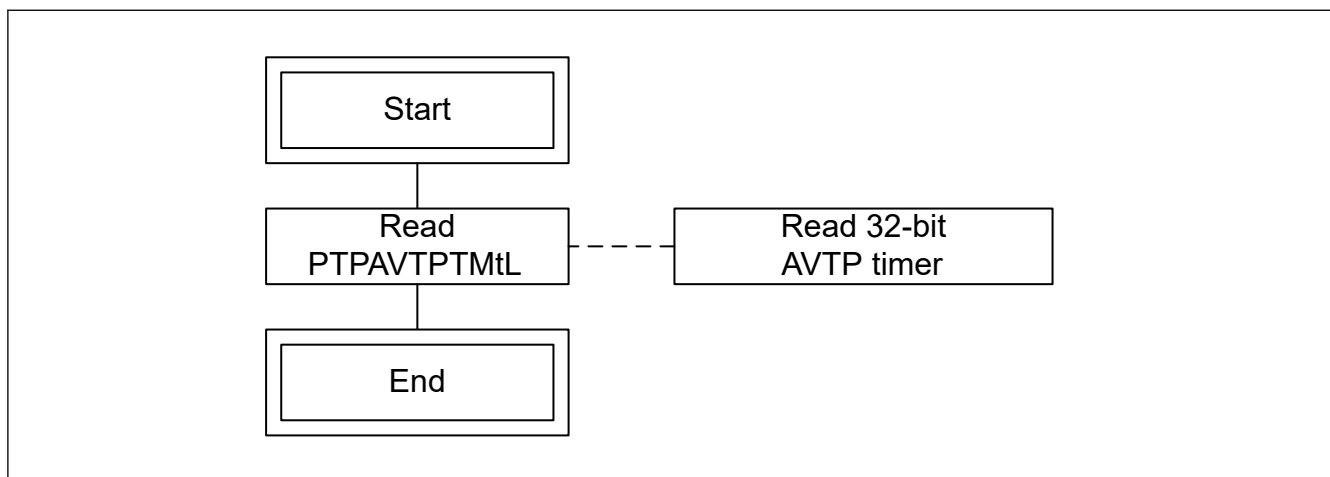


Figure 35.5 32-bit AVTP timer t reading flow

35.4.1.3.3 64-bit AVTP Timer t Reading Flow

Figure 35.6 describes the 64-bit AVTP timer t reading flow.

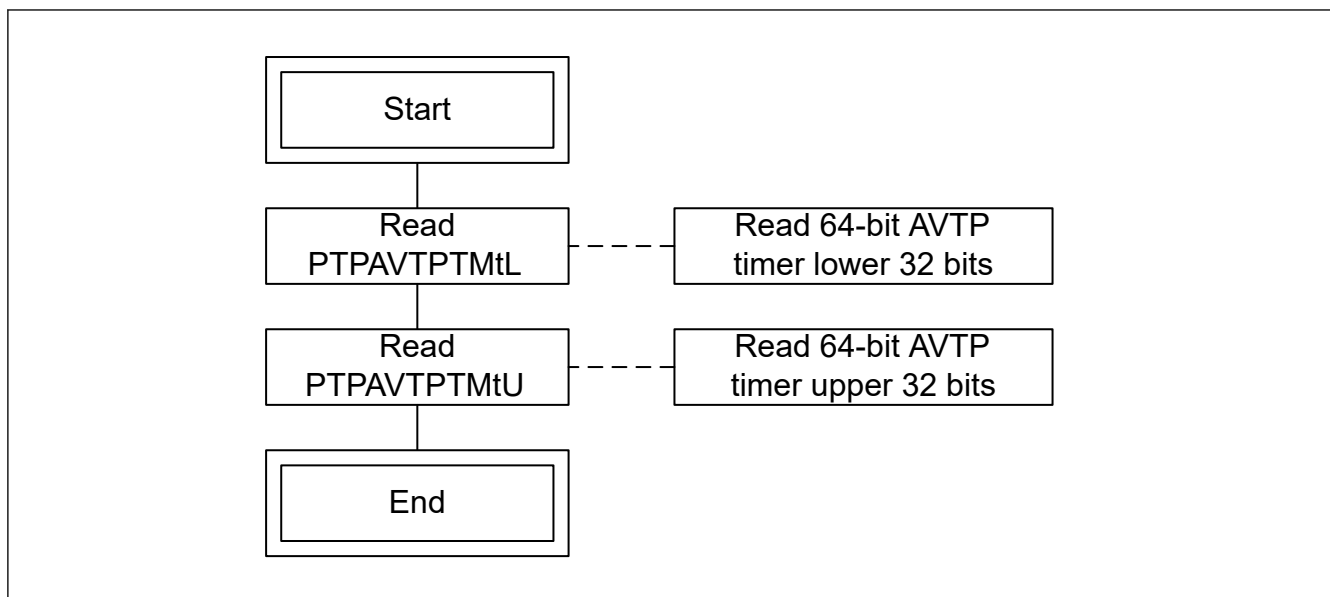


Figure 35.6 64-bit AVTP timer t reading flow

35.4.1.3.4 GPTP Timer t Reading Flow

Figure 35.7 describes the GPTP timer t reading flow.

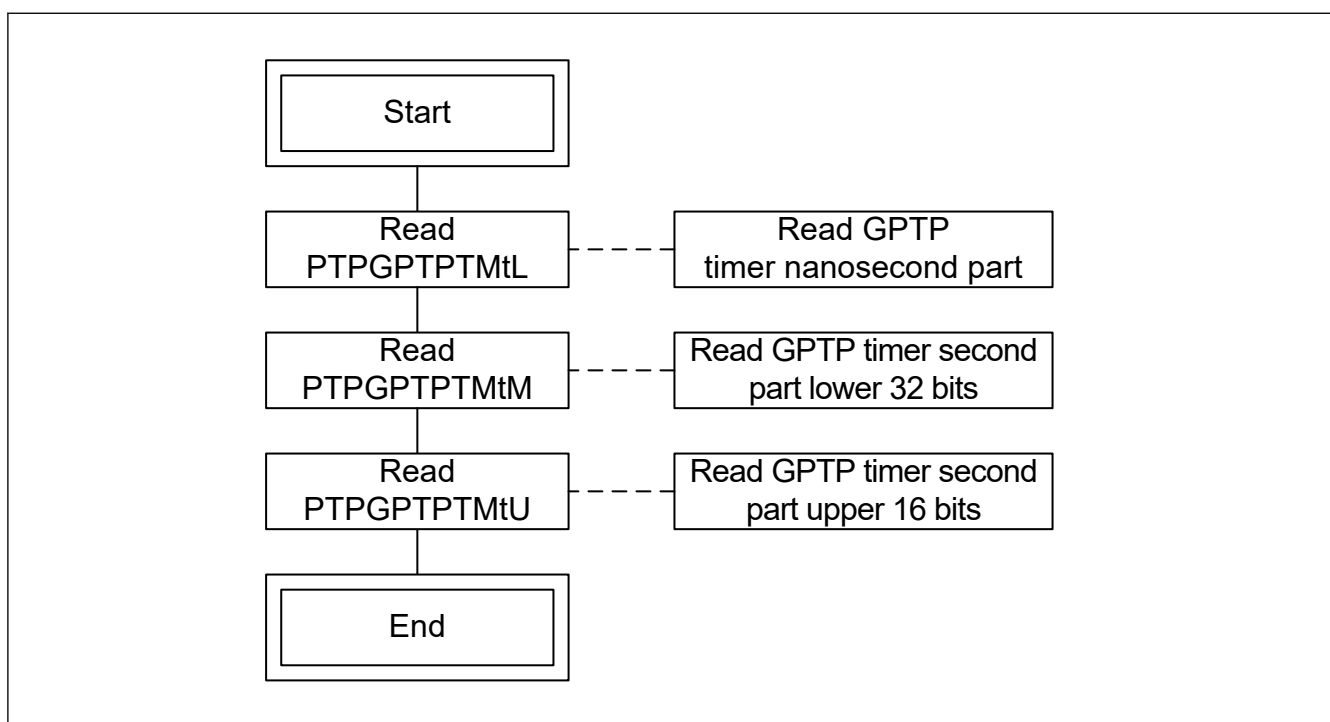


Figure 35.7 GPTP timer t reading flow

35.4.1.3.5 Register Writable Without Software Flow for Timer t

Table 35.4 describes the registers writable without software flow and in which mode they can be written.

Table 35.4 Writable without software flow registers (1 of 2)

(✓ = writable, × = not writable)

Register name	Writable anytime
PTPTMEC	✓

Table 35.4 Writable without software flow registers (2 of 2)

(✓ = writable, × = not writable)

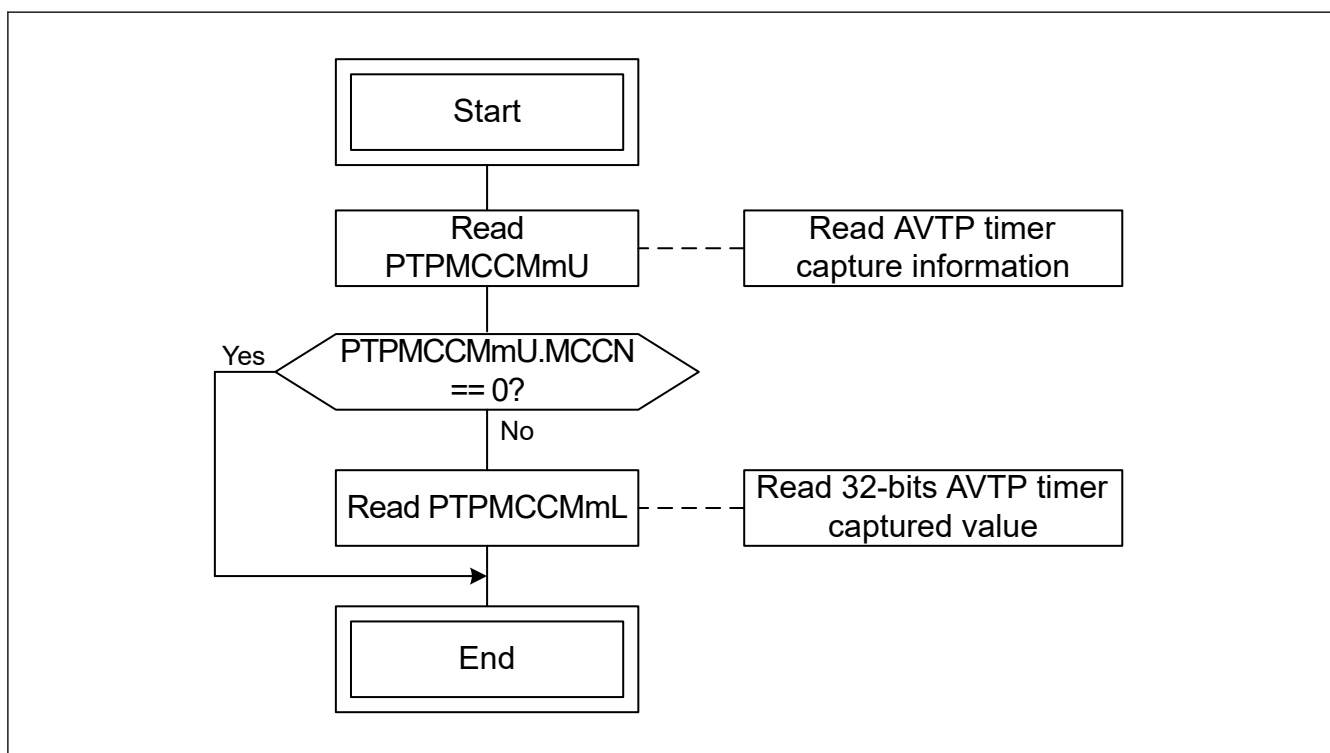
Register name	Writable anytime
PTPTMDC	✓
PTPTIVCt	✓

35.4.1.4 Media Clock m Capture Flows (m = 0, 1)

This section describes the flows to set and monitor media clock m capture.

35.4.1.4.1 32-bit AVTP Timer Captured Value Reading Flow

Figure 35.8 describes the 32-bit AVTP timer captured value reading flow.

**Figure 35.8** 32-bit AVTP timer captured value reading flow

35.4.1.4.2 64-bit AVTP Timer Captured Value Reading Flow

Figure 35.9 describes the 64-bit AVTP timer captured value reading flow.

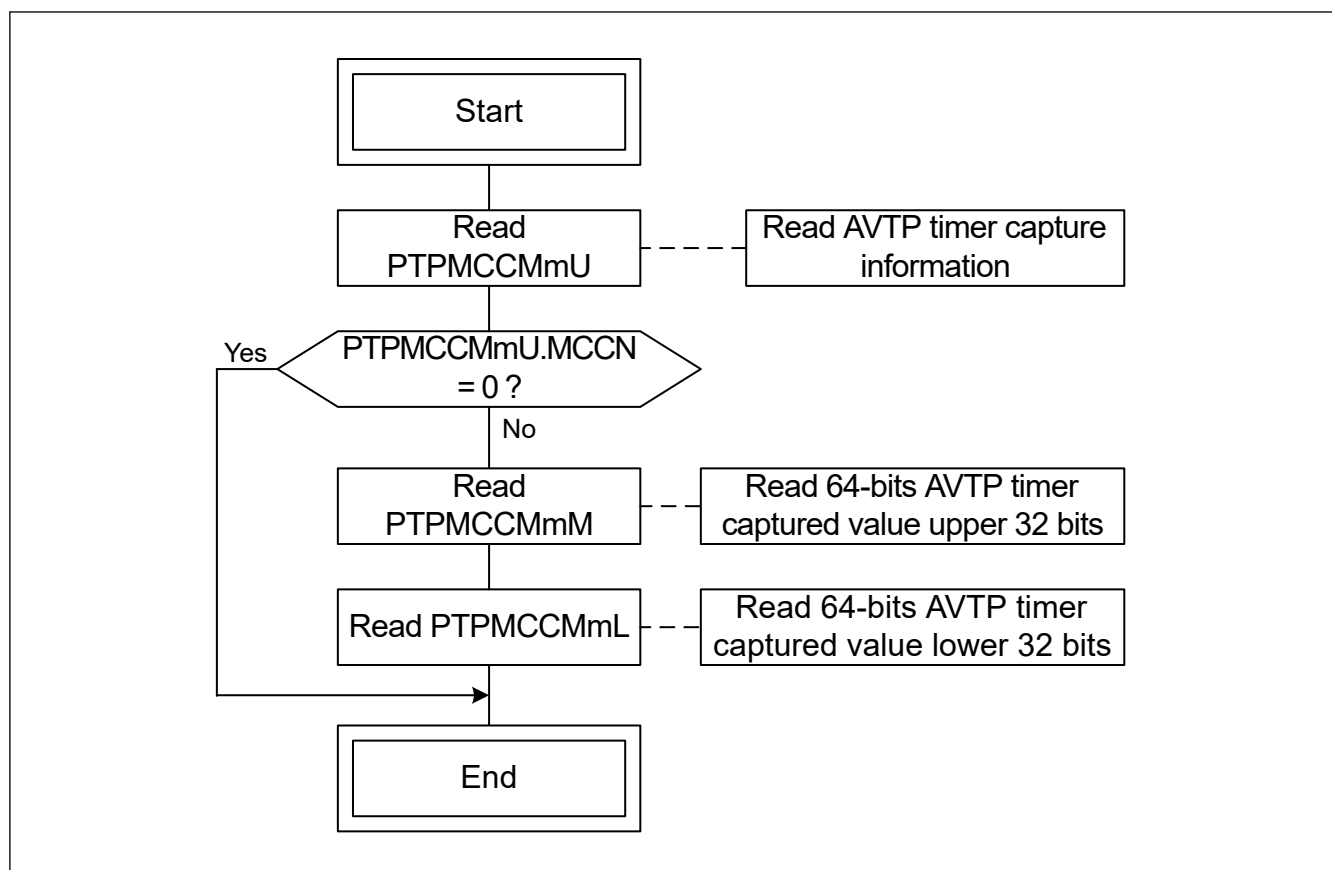


Figure 35.9 64-bit AVTP timer captured value reading flow

35.4.1.4.3 GPTP Timer Captured Value Reading Flow

Figure 35.10 describes the GPTP timer captured value reading flow.

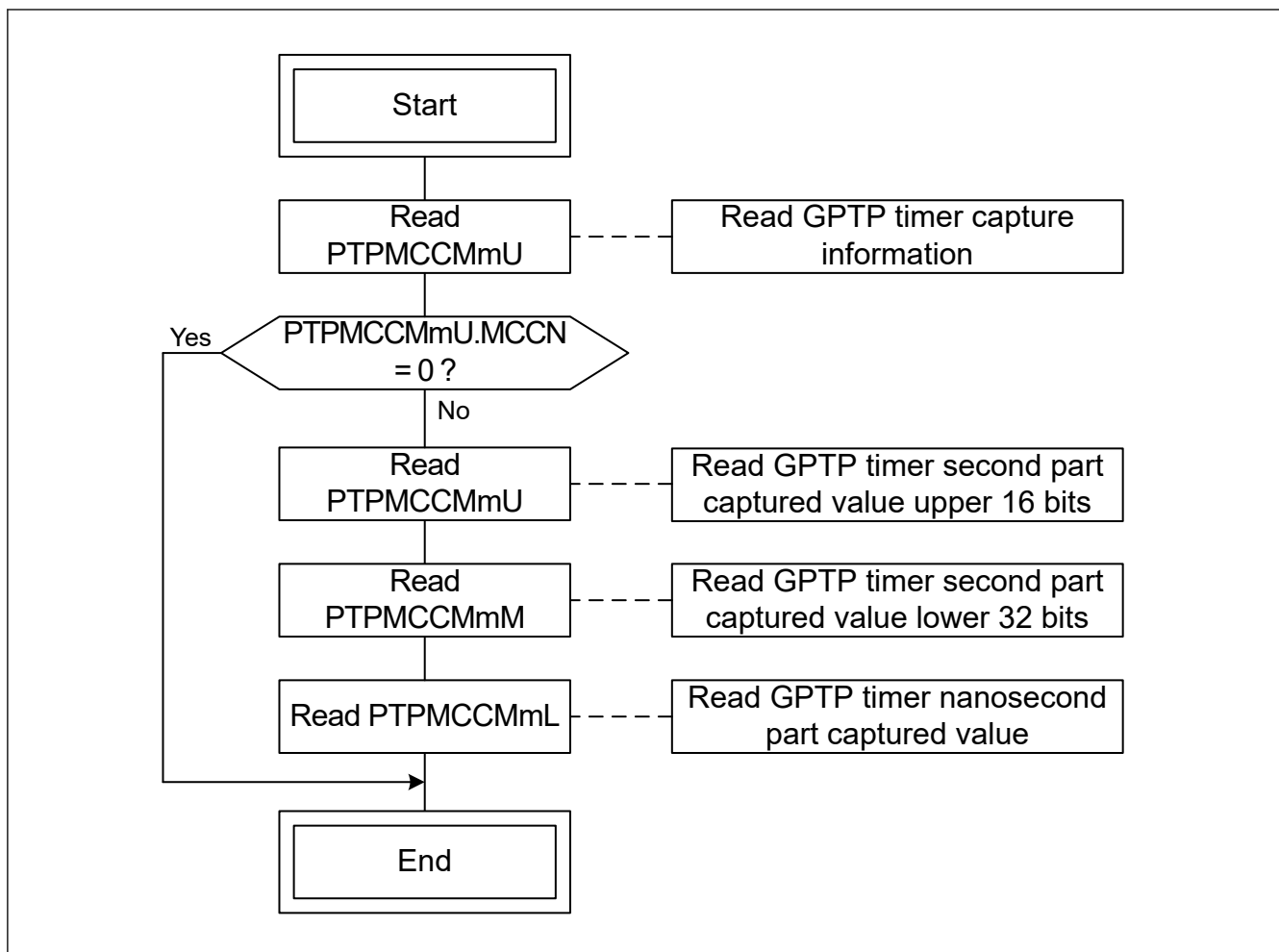


Figure 35.10 GPTP timer captured value reading flow

35.4.1.4.4 Register Writable Without Software Flow for Media Clock m Capture

Table 35.5 describes the registers writable without software flow and in which mode they can be written.

Table 35.5 Writable without software flow registers

(✓ = writable, × = not writable)

Register name	Writable anytime
PTPMCCm	✓

35.4.1.5 Media Clock m Recovery Flows (m = 0, 1)

This section describes the flows to set and monitor media clock m recovery.

35.4.1.5.1 Media Clock Recovery Time Value Setting Flow

Figure 35.11 describes the media clock recovery time value setting flow.

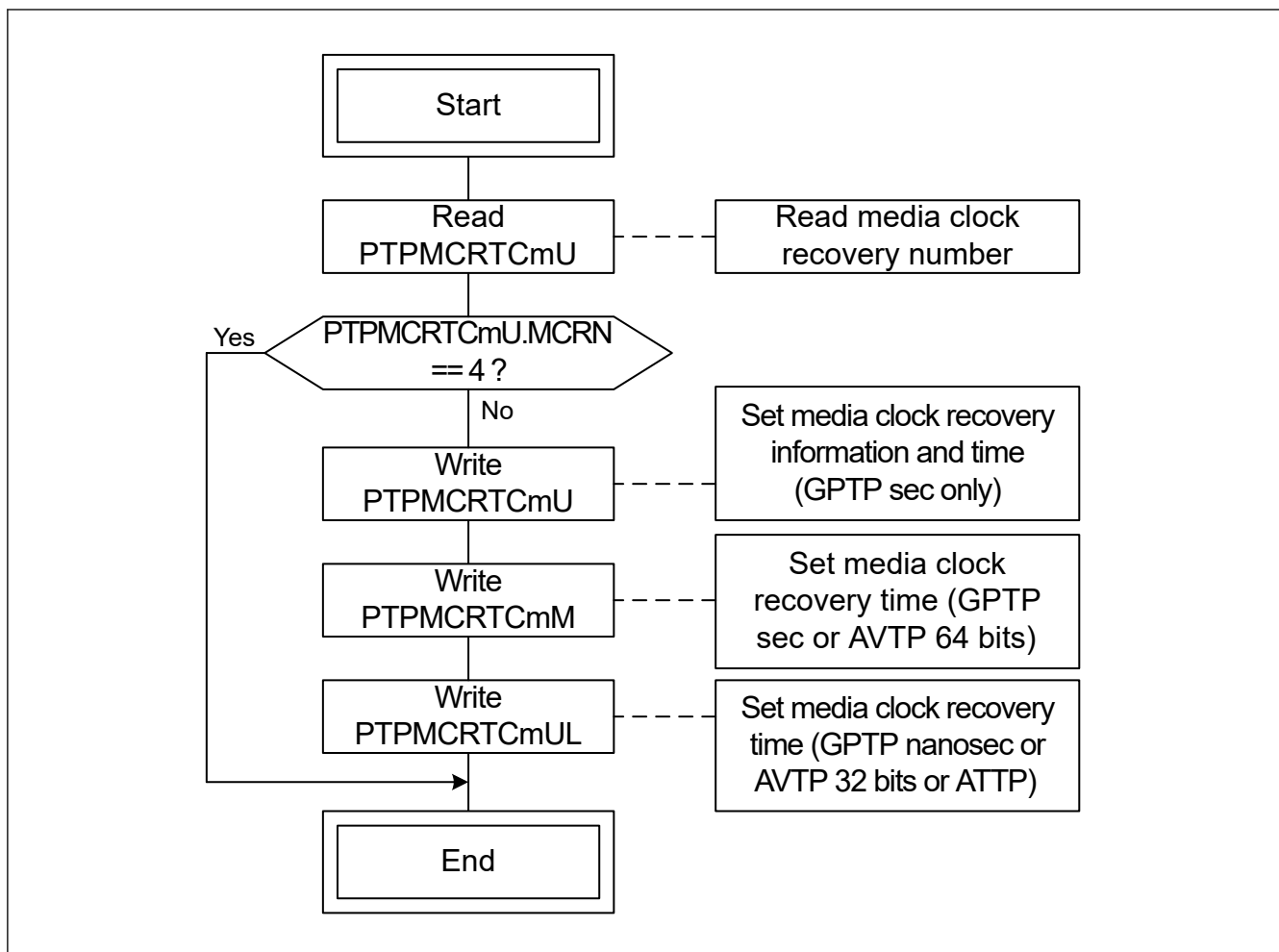


Figure 35.11 Media clock recovery time value setting flow

35.4.1.5.2 Media Clock Recovery Buffer Clear Flow

Figure 35.12 describes the media clock recovery buffer clear flow.

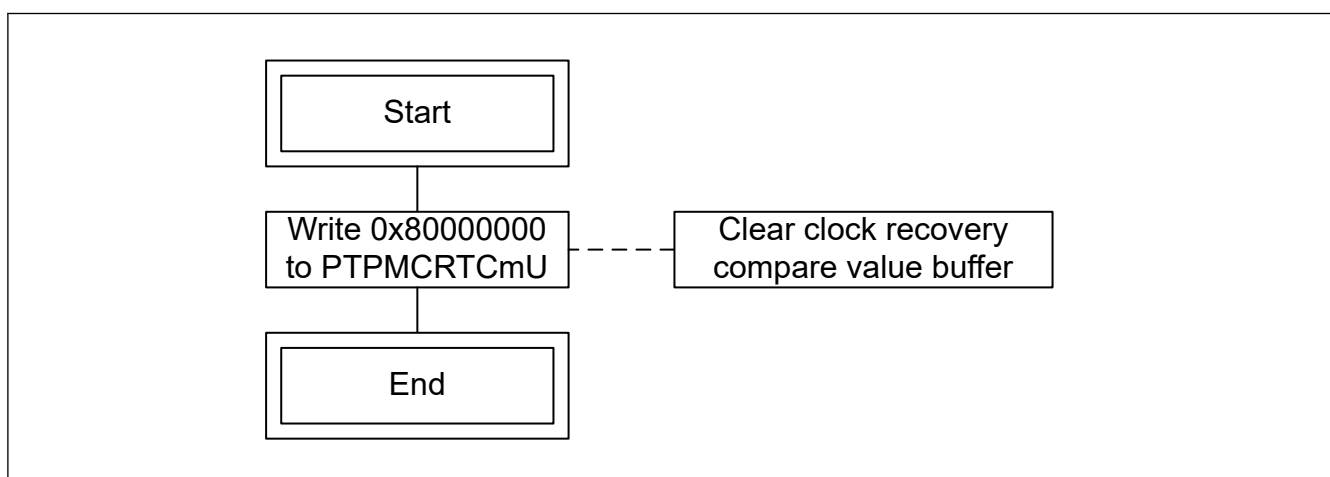


Figure 35.12 Media clock recovery buffer clear flow

35.4.1.5.3 Register Writable Without Software Flow for Media Clock m Recovery

Table 35.6 describes the registers writable without software flow and in which mode they can be written.

Table 35.6 Writable without software flow registers

(✓ = writable, × = not writable)

Register name	Writable anytime
PTPMCRCm	✓

35.4.1.6 Cyclic Compare c Flows (c = 0 to 7)

This section describes the flows to set and monitor cyclic compare c.

35.4.1.6.1 Register Writable Without Software Flow for Cyclic Compare c

Table 35.7 describes the registers writable without software flow and in which mode they can be written.

Table 35.7 Writable without software flow registers

(✓ = writable, × = not writable)

Register name	Writable anytime
PTPCCCc0	✓
PTPCCCc1	✓

35.4.1.7 Register Writable Without Software Flow

This section describes registers that have not been described. These registers can be changed dynamically. However, it is necessary that the initial settings such as the clock enabling have been completed.

35.5 Functional Description

35.5.1 Timer t (t = 0, 1) [gPTP]

Each timer instance includes three timers:

- A GPTP timer
- A 64-bit AVTP timer
- A 32-bit AVTP timer.

35.5.1.1 GPTP Timer t

The GPTP timer t GPTP[t] can be enabled by setting PTPTMEC.TE to 1. When enabled, GPTP timer t automatically calculates its value using equation (1).

$$(1) \text{ gtpTimer}[t] = \text{TimerOffset}[t] + \text{FreeRunningTimerValue}[t]$$

Where:

- gtpTimer[t] is the calculated timer value GPTP[t] and is divided in two parts. The second part gtpSecond[t] is GPTP[t][77:30] and the nanosecond part gtpNano[t] is GPTP[t][29:0].
- TimerOffset[t] is {PTPTOVcU.TOVU, PTPTOVcM.TOVM, PTPTOVcL.TOVL}
- FreeRunningTimerValue[t] is the value of a free running timer starting from 0 (when timer is disabled) and is incremented by PTPTIVcT.TIV every cycle (the free running timer has a 27-bit sub-nanosecond part not described here so it should be set to $\text{clk_period}[\text{ns}] \times 2^{27}$ where clk_period is clock clk period).

Functions:

- When GPTP timer t is a master, PTPTIVcT.TIV and {PTPTOVcU.TOVU, PTPTOVcM.TOVM, PTPTOVcL.TOVL} registers can be kept to fixed values.
- When GPTP timer t is a slave, PTPTIVcT.TIV register can be used in priority for time resynchronization. If GPTP timer t exceeds the corresponding master value, {PTPTOVcU.TOVU, PTPTOVcM.TOVM, PTPTOVcL.TOVL} registers can be used for time resynchronization.

35.5.1.2 64-bit AVTP Timer t

The 64-bit AVTP timer t AVTP64[t] is used as a reference to synchronize any necessary media clocks and to determine when the payload of a stream is to be presented to the time-sensitive application. Media clocks vary with audio/video types therefore the exact usage of the AVTP presentation time is media-format dependent [1722].

The 64-bit AVTP timer t is generated from the GPTP timer t and can be calculated using equation (2).

$$(2) \text{avtpTimer64}[t] = (\text{gtpSecond}[t] \times 10^9 + \text{gtpNano}[t]) \bmod 2^{64}$$

Where:

- avtpTimer64[t] is the calculated timer value AVTP64[t]
- gtpSecond[t] is gtpTimer[t] second part GPTP[t][77:30]
- gtpNano[t] is gtpTimer[t] nanosecond part GPTP[t][29:0].

35.5.1.3 32-bit AVTP Timer t

The 32-bit AVTP timer t AVTP32[t] is the 32-bit version of 64-bit AVTP timer t and can be calculated using equation (3).

$$(3) \text{avtpTimer32}[t] = (\text{gtpSecond}[t] \times 10^9 + \text{gtpNano}[t]) \bmod 2^{32}$$

Where:

- avtpTimer32[t] is the calculated timer value AVTP32[t] (AVTP64[t][31:0])
- gtpSecond[t] is gtpTimer[t] second part GPTP[t][77:30]
- gtpNano[t] is gtpTimer[t] nanosecond part GPTP[t][29:0].

35.5.1.4 PPS Signal t

The PPS[t] signal is used to check if GPTP synchronization is successful. As per GPTP standard [PTP], PPS[t] signal should be output as pins at the chip level to allow synchronization checking on oscilloscope. PPS[t] is the PPS signal corresponding to timer number t. For PPS signal details, see the GPTP standard [gPTP].

35.5.2 Media Clock m Capture (m = 0, 1)

Media clock m capture aims at capturing timer values on MEDIA_IN[m] positive edge and/or negative edge.

Functions:

- PTPMCCCm.MCPEE is used to enable timer value capture on MEDIA_IN[m] positive edge
- PTPMCCCm.MCNEE is used to enable timer value capture on MEDIA_IN[m] negative edge
- PTPMCCCm.MCTNS is used to select timer number for timer value capture
- PTPMCCCm.MCTTS is used to select between GPTP and AVTP timer for timer value capture
- PTPMCCMmL/mM/mU are used to read the captured timer values
- PTPIS0.MCCSn interrupt is used to signal to the CPU that a timer value has been captured
- PTPIS0.MCCOESn interrupt is used to signal to the CPU that a captured timer value has been lost because the timer capture value FIFO was full.

35.5.3 Media Clock m Recovery (m = 0, 1)

Media clock m recovery aims at creating an internal clock mediaClock[m] which can be mapped to an output pin, (see [section 35.5.4. Media Clock m Mapping to Output Pins](#)).

Functions:

- PTPMCRCm.MRTNS is used to select timer number for clock recovery
- PTPMCRCm.MRTTS is used to select between GPTP and AVTP timers for clock recovery
- PTPMCRCm.MRAMS is used to select between 32-bit AVTP timer and 64-bit AVTP timer for clock recovery when PTPMCRCm.MRTTS is set to AVTP timer
- PTPMCRCmL/mM/mU are used to add a timer value and command to execute to mediaClock[m]

- PTPIS0.MCRMS[m] interrupt is used to signal to the CPU that the oldest timer value (which has not matched yet) set in PTPMCRCmL/mM/mU matched the selected timer therefore mediaClock[m] value has been updated.

Figure 35.13 describes mediaClock[m] behavior when a timer value matches depending on the timer value type setting and the initial value of mediaClock[m].

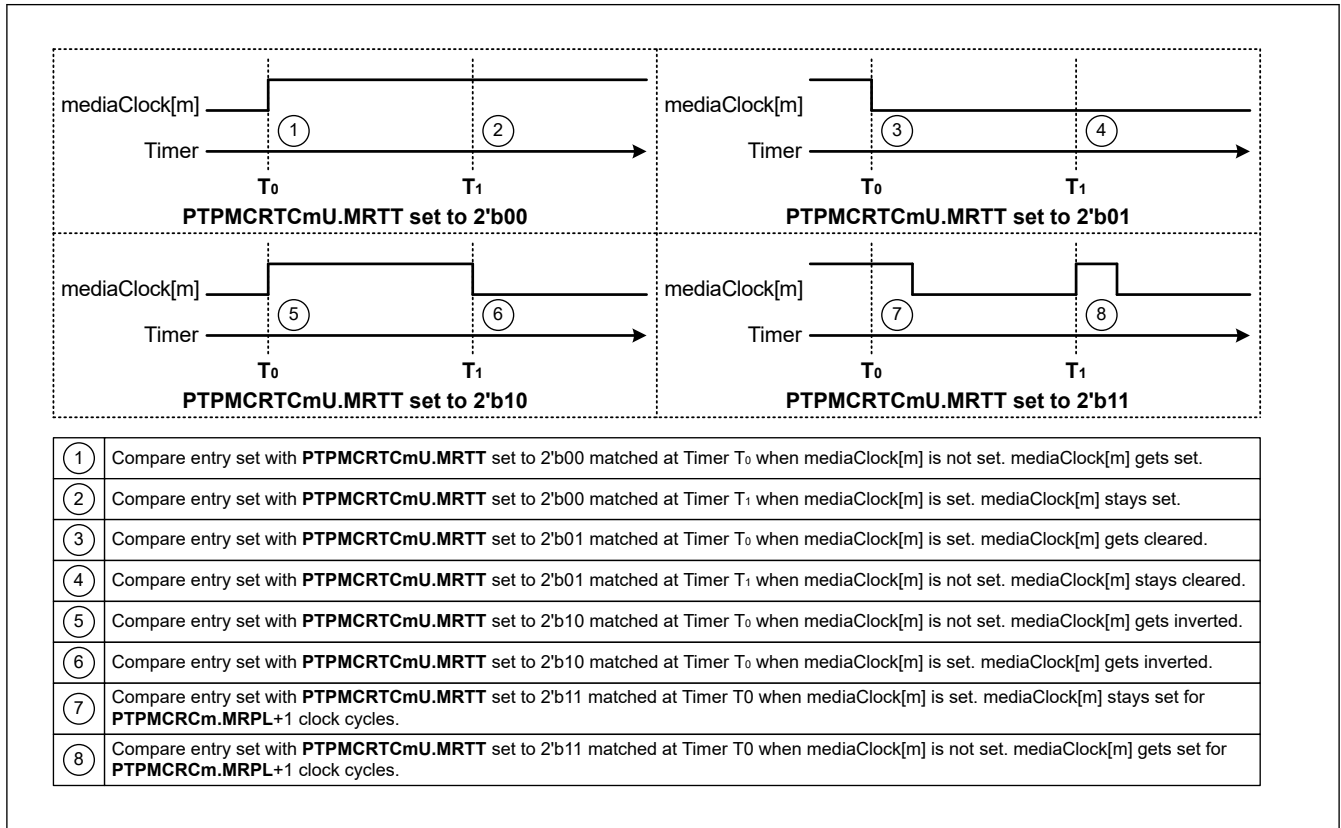


Figure 35.13 mediaClock[m] behavior

Note: When timer matched an entry with PTPMCRTCmU.MRTT set to 2'b11 and the next match occurs before mediaClock[m] deassertion, mediaClock[m] deassertion command is overwritten and does not occur.

35.5.4 Media Clock m Mapping to Output Pins

The mediaClock[m] signal created by media clock m recovery (section 35.5.3. Media Clock m Recovery ($m = 0, 1$)) can be output to be used inside the chip or as a chip pin.

Functions:

- By setting PTPMCPCm.PE to 1 and PTPMCPCm.MRS to m, mediaClock[m] signal is assigned to MEDIA_OUT[p] output pin.

35.5.5 Cyclic Compare c ($c = 0$ to 7)

Cyclic compare c aims at creating a pulse cyclically on CYCLIC_COMP[i][c] for $i = 0, 1$ interface to trigger an event. It can be used, for example, to trigger a periodic data transmission for GPTP synchronization (network control frame transmission).

Functions:

- PTPCCCc0.CCTNS is used to select timer number for cyclic compare. During cyclic compare, timer offset is ignored, and only incremental value is taken in account.
- PTPCCCc1.CCV is used to enable cycle compare. When cycle compare c is enabled, a pulse is sent every PTPCCCc1.CCV to CYCLIC_COMP[PTPCCCc0.CCOPS][c].
- When this feature is enabled (when the PTPCCCc1.CCV configuration is updated to a value greater than 32), the first pulse from comparator c is output immediately (after one system clock), regardless of PTPCCCc1.CCV configuration.

35.6 Pulse Output Timer Module (plsotim)

35.6.1 Overview of plsotim

The pulse output timer module (plsotim), a submodule of gPTP, produces a pulsed output according to the base period of a gPTP timer.

Table 35.8 lists the plsotim specifications.

Table 35.8 plsotim specifications

Item	Function	Description
Overall function	Pulse generator	Pulse generation timer, pulsed output, and interrupt output.
Data provision		- Pulsed output channel: 4 ch The pulse cycle: 80 bits ([79:32] = second and [31:0] = ns) Pulse width: 20 bits ([19:0] = counter) - Interruption output channel: 4 ch.
Functions	PTP timer select	The PTP timer to be used as the reference time can be selected.
	PTP_PULSE_GEN	Uses the reference time of the GPTP timer to generate pulse outputs of any period and any pulse width.
	Interrupt pulse output	The rising edge of the pulse output is detected, and the interrupt pulse is output.

35.6.2 Plsotim Register List

Table 35.9 Plsotim registers

Offset address	Register name	Register symbol	Write access mode
0x1000	Pulse Output Timer Configuration Register	POTCFGR	—
0x1004 + 0x30 × n	Pulse Output Timer Control Register n (n = 0 to 3)	POTCRn	—
0x1008 + 0x30 × n	Pulse Output Start Time Setting Register nU (n = 0 to 3)	POTSTRnU	—
0x100C + 0x30 × n	Pulse Output Start Time Setting Register nM (n = 0 to 3)	POTSTRnM	—
0x1010 + 0x30 × n	Pulse Output Start Time Setting Register nL (n = 0 to 3)	POTSTRnL	—
0x1014 + 0x30 × n	Period Setting Register nU (n = 0 to 3)	POTPERnU	—
0x1018 + 0x30 × n	Period Setting Register nM (n = 0 to 3)	POTPERnM	—
0x101C + 0x30 × n	Period Setting Register nL (n = 0 to 3)	POTPERnL	—
0x1020 + 0x30 × n	Pulse Width Setting Register n (n = 0 to 3)	POTPWRn	—
0x1028 + 0x30 × n	Time Capture Register nU (n = 0 to 3)	POTCPRnU	—
0x102C + 0x30 × n	Time Capture Register nM (n = 0 to 3)	POTCPRnM	—
0x1030 + 0x30 × n	Time Capture Register nL (n = 0 to 3)	POTCPRnL	—

Note: Base address: GPTP = 0x403E_0000, GPTP_NS = 0x503E_0000

35.6.3 Plsotim Register Descriptions

35.6.3.1 POTCFGR : Pulse Output Timer Configuration Register

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	REFSEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	REFSEL	Reference Timer Select 0: Select and use the reference time of GPTP timer 0 1: Select and use the reference time of GPTP timer 1	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.2 POTCRn : Pulse Output Timer Control Register n (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1004 + 0x30 × n

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	START
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

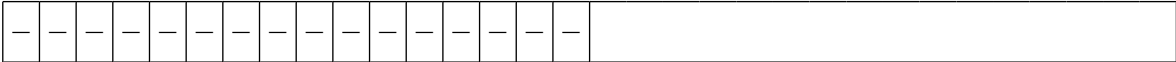
Bit	Symbol	Function	R/W
0	START	Pulse Output Enable Pulse generation is performed by taking in the set value of the pulse generation register when this register becomes 1. If set to 0, no pulse is output. By writing 0 during pulse output, pulse output stops (0 output). 0: Pulse output stop 1: Pulse output permission	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.3 POTSTRnU : Pulse Output Start Time Setting Register nU (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1008 + 0x30 × n

Bit position: 31 15 0

Bit field: 

Value after reset: 0

Bit	Symbol	Function	R/W
15:0	n/a	Seconds (upper 16 bits) of pulse output start time. Functions: - Sets the lower 16 bits second of the time to start pulse output - For comparison with the reference time of the GPTP timer. Valid value - The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 - Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.4 POTSTRnM : Pulse Output Start Time Setting Register nM (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x100C + 0x30 × n

Bit position: 31 0

Bit field: 

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	n/a	Seconds (upper 32 bits) of pulse output start time. Functions: - Sets the lower 32 bits second of the time to start pulse output - For comparison with the reference time of the GPTP timer. Valid value - The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 - Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation.	R/W

35.6.3.5 POTSTRnL : Pulse Output Start Time Setting Register nL (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1010 + 0x30 × n

Bit position: 31 29 0

Bit field: 

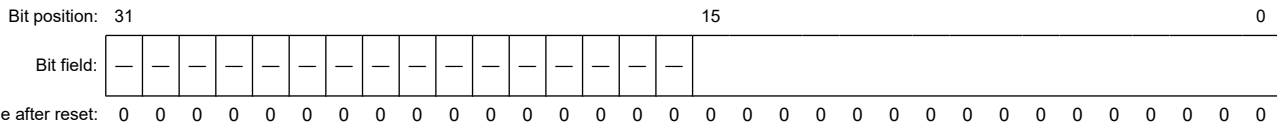
Value after reset: 0

Bit	Symbol	Function	R/W
29:0	n/a	Nanoseconds (29 bits) of pulse output start time. Functions: - Sets the nanosecond of the time to start pulse output - For comparison with the reference time of the GPTP timer. Valid value - The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 - Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation. Restrictions: - Set the value between 0 and 0x3B9AC9FF. 0x3B9ACA00 is 1 second, therefore set using the Second register.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.6 POTPERnU : Period Setting Register nU (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1014 + 0x30 × n

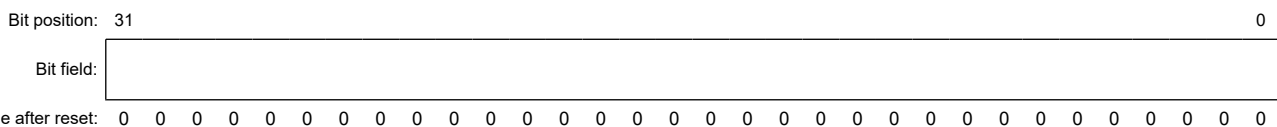


Bit	Symbol	Function	R/W
15:0	n/a	Seconds (upper 16 bits) of pulse output period. Functions: - Sets the lower 16 bit seconds of the time to start pulse output - Calculates the reference time of the start time to be compared next. Valid value - The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 - Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.7 POTPERnM : Period Setting Register nM (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1018 + 0x30 × n



Bit	Symbol	Function	R/W
31:0	n/a	Seconds (upper 32 bits) of pulse output period. Functions: - Sets the lower 32-bit seconds period of the pulse output - Calculates the reference time of the start time to be compared next. Valid value: - The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 - Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation.	R/W

35.6.3.8 POTPERnL : Period Setting Register nL (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x101C + 0x30 × n

Bit position: 31 29

0

Bit field:

—	—																												
---	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Value after reset: 0

Bit	Symbol	Function	R/W
29:0	n/a	Nanoseconds (30 bits) of pulse output period. Functions: • Sets the nanosecond of the time-to-period pulse output • Calculates the reference time of the start time to be compared next. Valid value: • The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 • Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation • When reading, the written value can be read. Restrictions: • Set the value between 0 and 0x3B9AC9FF. 0x3B9ACA00 is 1 second, therefore set using the Second register.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.9 POTPWRn : Pulse Width Setting Register n (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1020 + 0x30 × n

Bit position: 31

15

0

Bit field:

—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—															
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

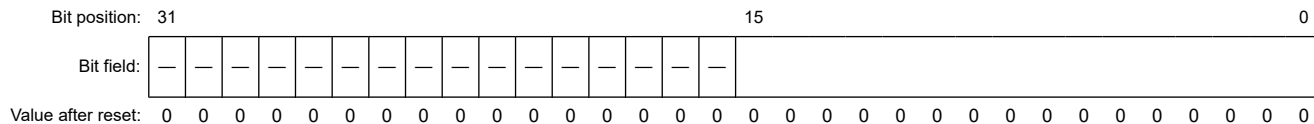
Value after reset: 0

Bit	Symbol	Function	R/W
15:0	n/a	High Pulse Width Count Number for the Pulse Output. Functions: • Sets the number of counts for the high pulse width • Counting begins from the start of pulse output • The high pulse width time varies depending on the clock period. Valid value: • The register value to generate the pulse output uses the value when the POTCRn.START bit changes from 0 to 1 • Even if the register value is changed after the POTCRn.START bit becomes 1, it is not reflected in the pulse generation.	R/W
31:16	—	These bits are read as 0. The write value should be 0.	R/W

35.6.3.10 POTCPRnU : Time Capture Register nU (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1028 + 0x30 × n

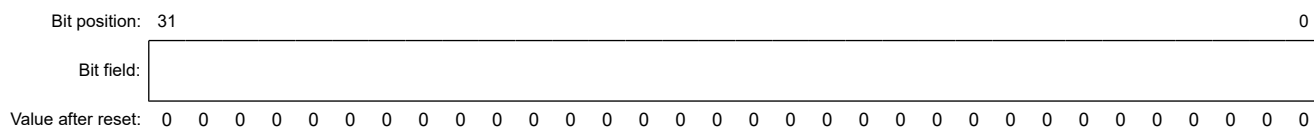


Bit	Symbol	Function	R/W
15:0	n/a	The upper 16-bit value of seconds for the pulse output start time.	R
31:16	—	These bits are read as 0.	R

35.6.3.11 POTCPRnM : Time Capture Register nM (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x102C + 0x30 × n

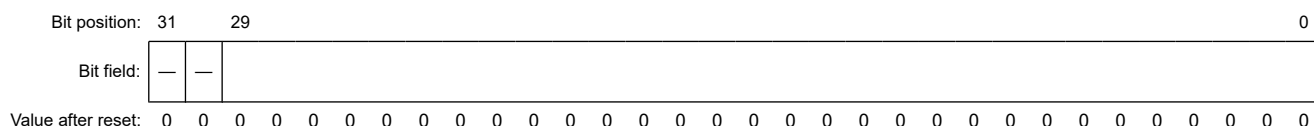


Bit	Symbol	Function	R/W
31:0	n/a	The lower 32-bit value of seconds for the pulse output start time.	R

35.6.3.12 POTCPRnL : Time Capture Register nL (n = 0 to 3)

Base address: GPTP = 0x403E_0000
GPTP_NS = 0x503E_0000

Offset address: 0x1030 + 0x30 × n



Bit	Symbol	Function	R/W
29:0	n/a	The nanosecond value for the pulse output start time.	R
31:30	—	These bits are read as 0.	R

35.6.4 Operation of plstotim

35.6.4.1 Pulse Output Timer Operation Settings

Figure 35.14 shows the operation setting flow of the pulse output timer n.

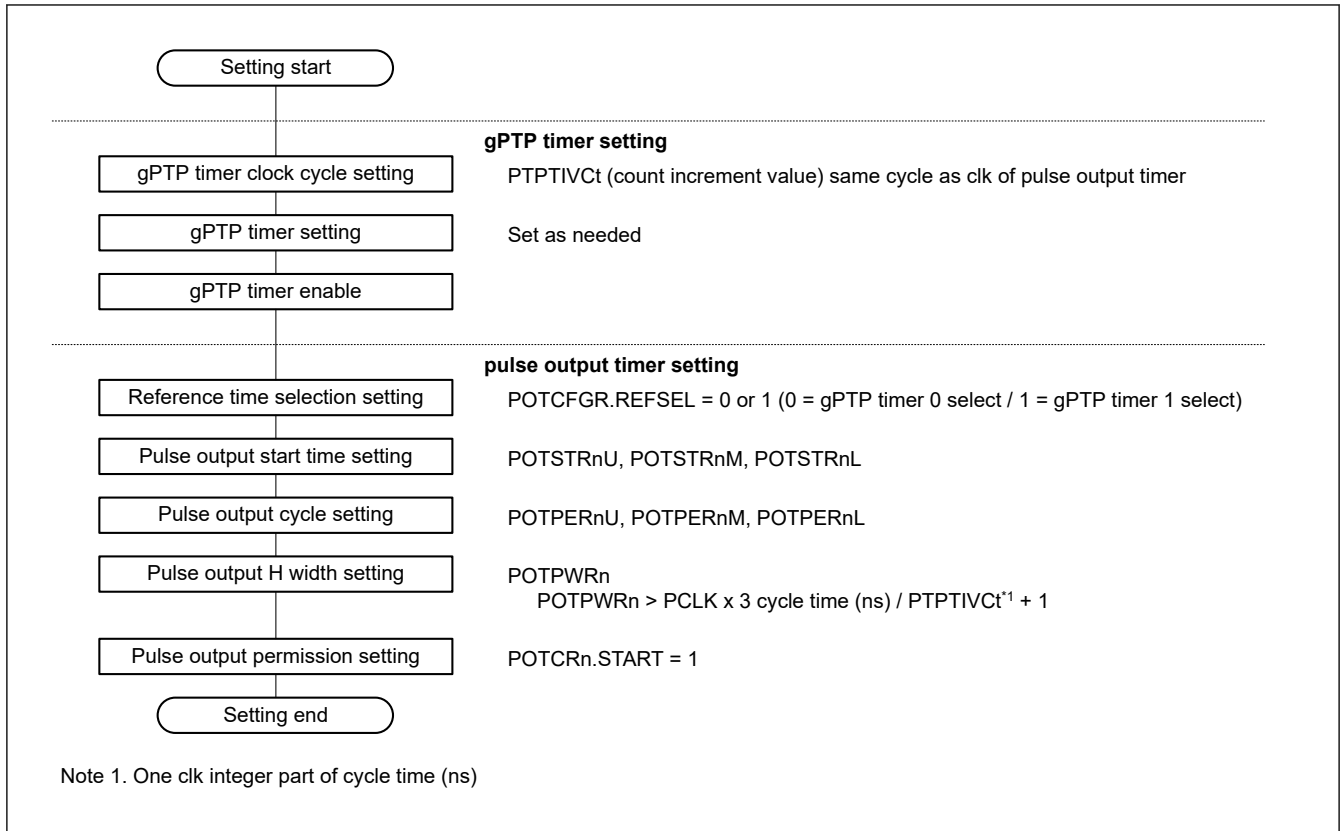


Figure 35.14 Operation setting flow of the pulse output timer

35.6.4.2 Pulse Output Timer n (n = 0 to 3)

This module uses the reference time of the GPTP timer to generate a pulse output of any period and any pulse width.

35.6.4.2.1 GPTP Timer Select

Select the GPTP timer of the reference time to be used by setting the POTCFGR register.

POTCFGR.REFSEL bit setting:

- 0 : Generates a pulse output using the reference time of GPTP timer 0.
- 1 : Generates a pulse output using the reference time of GPTP timer 1.

Functions:

- The reference time used for pulse generation can be selected by setting the POTCFGR register
- When POTCFGR.REFSEL = 0: Generates a pulse using the time input signal of GPTP timer 0
- When POTCFGR.REFSEL = 1: Generate a pulse using the time input signal of GPTP timer 1
- When changing the POTCFGR register, stop the pulse output (POTCRn.START = 0) before modification.

35.6.4.2.2 Pulse Output Generation Register

By setting the following registers, it is possible to generate a pulse with an arbitrary start time, an arbitrary period, and an arbitrary H pulse width.

- Pulse output start time register: POTSTRnU, POTSTRnM, POTSTRnL
- Pulse output cycle register: POTPERnU, POTPERnM, POTPERnL
- Pulse output H width register: POTPWRn.

When the H width register of the pulse output is 0, the pulse is not generated, so the L output is fixed.

If the pulse output cycle is set to be less than or equal to the H width time of the pulse output, the H output is fixed.

35.6.4.2.3 Pulse Output Generation

By changing the POTCRn.START bit from 0 to 1, the value of the pulse output generation register is captured and held.

Pulse is generated according to the set value of the held pulse output generation.

The output pulse is as follows.

Rise of the first pulse:

Pulse output H width register $\neq 0$: It becomes H at the set start time.

Pulse output H width register = 0: L output remains (no pulse output).

Pulse fall:

Pulse output cycle register value $>$ Pulse output H width register value $\times$ clk cycle.

When the clk number is counted by the amount of the H width register value after the output becomes H, it becomes L.

Pulse output cycle register value $\leq$ H width register value of pulse output $\times$ clk cycle.

The output remains H (H output is fixed until the POTCRn.START bit is set to 0).

Rise from the second time onward:

When H width register = other than 0 and pulse output cycle $>$ H width register value $\times$ clk cycle.

It becomes H at the time of the previous start time + pulse output cycle register value.

35.6.4.2.4 Pulse Output Stop

Pulse output can be stopped by setting the POTCRn.START bit to 0.

When stopped, the output is fixed at L.

When the POTCRn.START bit is set to 1 again, the register setting value is taken in and held, and a new pulse is generated and output.

Note: Even if the pulse output is stopped in the middle, it is not restarted. A new pulse is generated and output.

35.6.4.2.5 Timing of Pulse Output Permission

If the POTCRn.START bit is changed from 0 to 1 after the set pulse output start time, no pulse output is performed.

The pulse is output from the time when the reference time goes around once and the start time is reached again.

35.6.4.3 Pulse Output Timing Diagrams

Six examples of pulse outputs are shown in the timing diagrams as following:

1. [Figure 35.15](#) When changing the POTCRn.START bit from 0 to 1 after setting the register and before the start time
2. [Figure 35.16](#) When changing the POTCRn.START bit from 0 to 1 after setting the register and after the start time
3. [Figure 35.17](#) Pulse output H width register $\geq$ when the POTCRn.START bit is changed from 0 to 1 by setting the pulse output cycle
4. [Figure 35.18](#) When changing the POTCRn.START bit from 0 to 1 with pulse output H width register = 0
5. [Figure 35.19](#) When the pulse output is stopped with the POTCRn.START bit set to 0
6. [Figure 35.20](#) When the register setting is changed with the POTCRn.START bit set to 1.

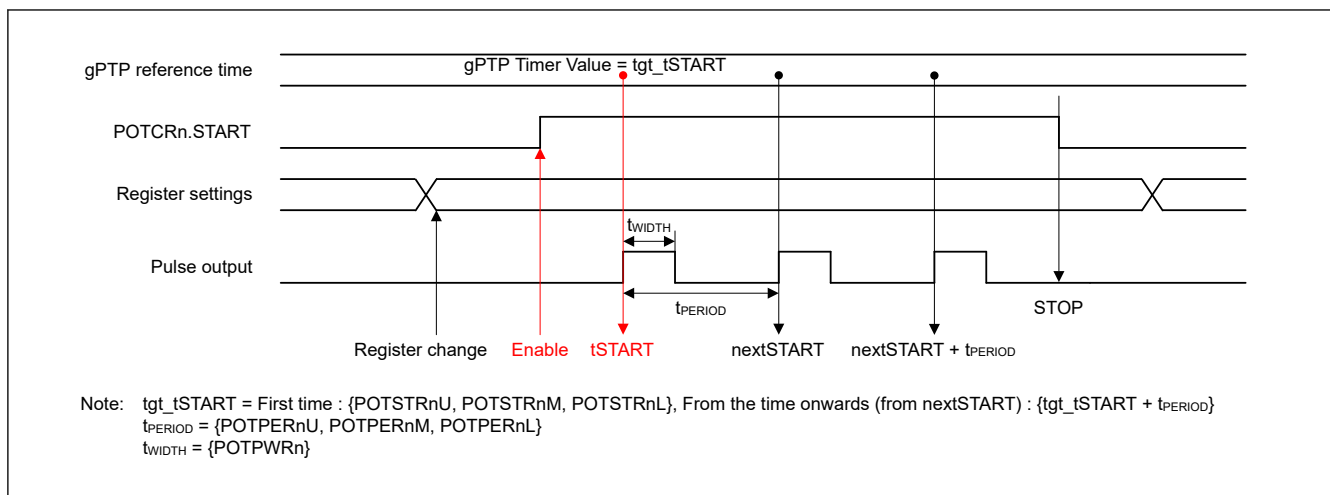


Figure 35.15 When changing the POTCRn.START bit from 0 to 1 after setting the register and before the start time

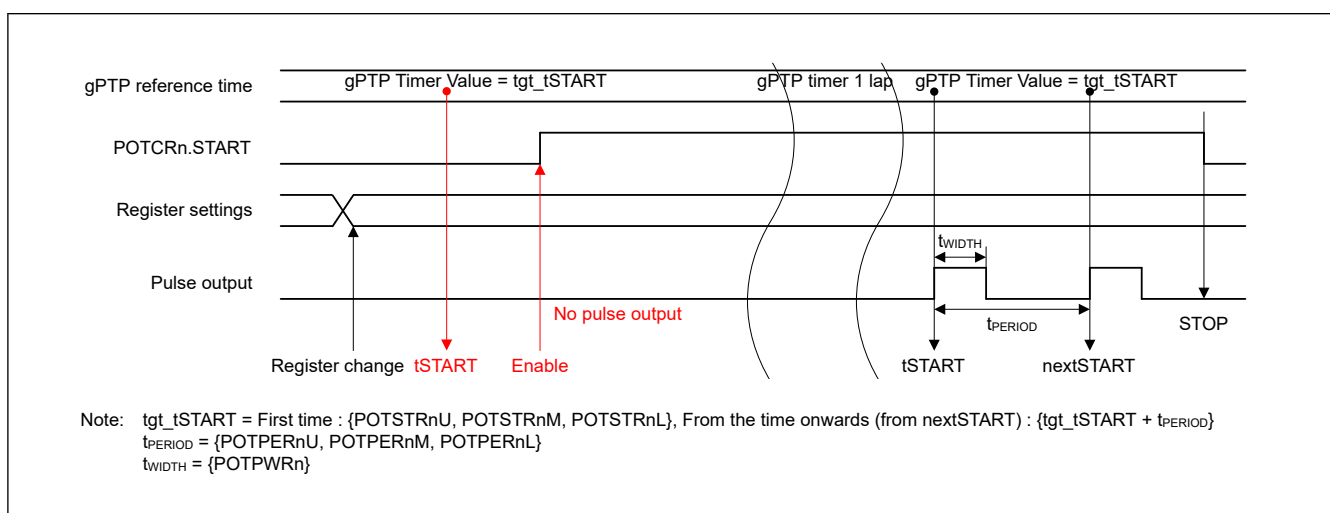


Figure 35.16 When changing the POTCRn.START bit from 0 to 1 after setting the register and after the start time

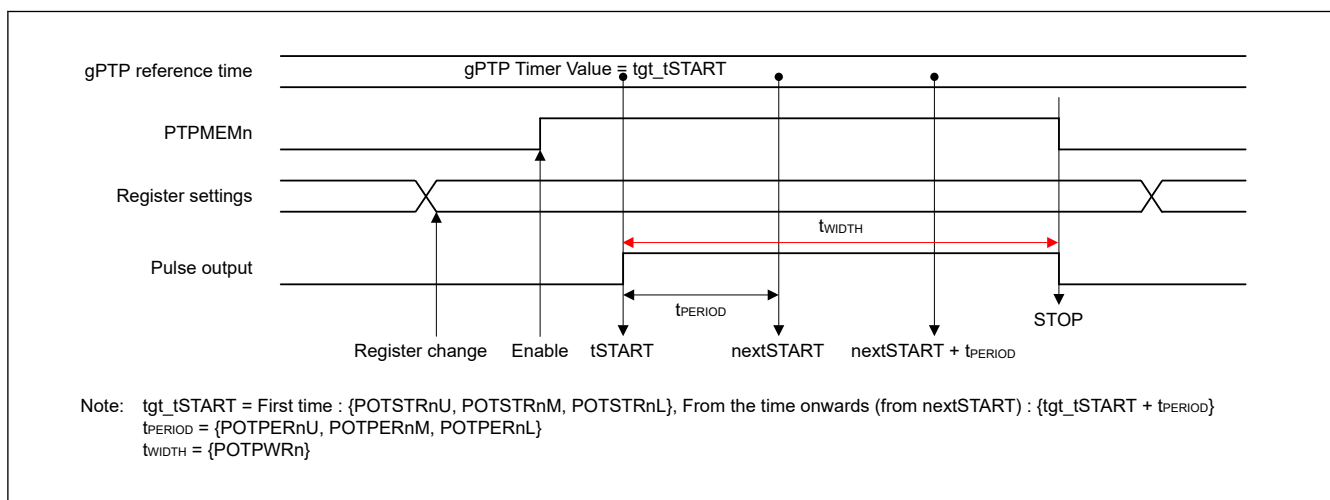


Figure 35.17 Pulse output H width register $\geq$ when the POTCRn.START bit is changed from 0 to 1 by setting the pulse output cycle

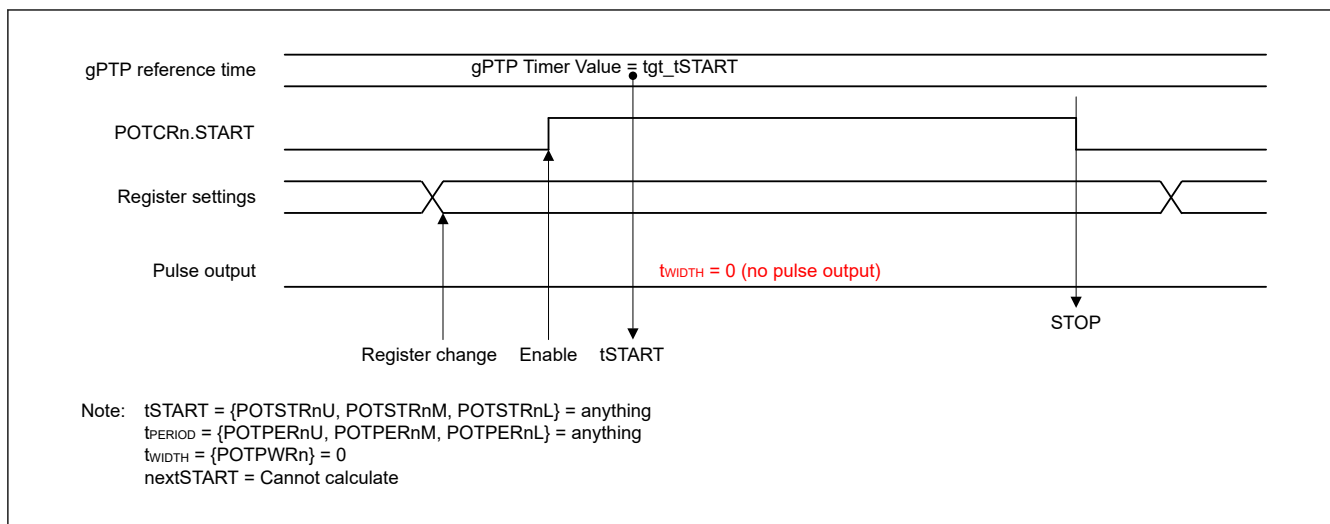


Figure 35.18 When changing the POTCRn.START bit from 0 to 1 with pulse output H width register = 0

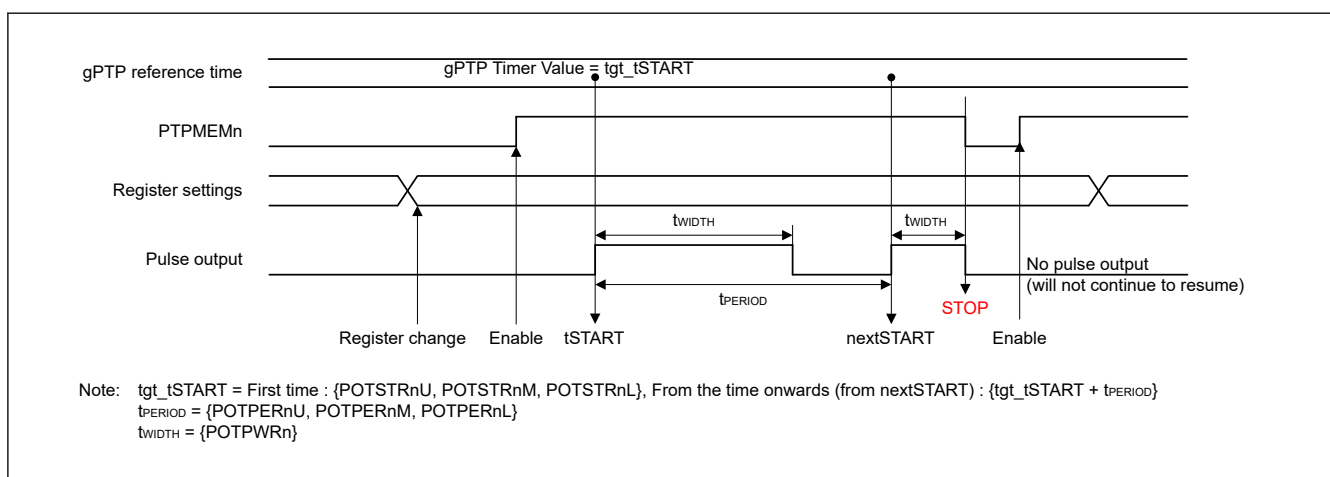


Figure 35.19 When the pulse output is stopped with the POTCRn.START bit set to 0

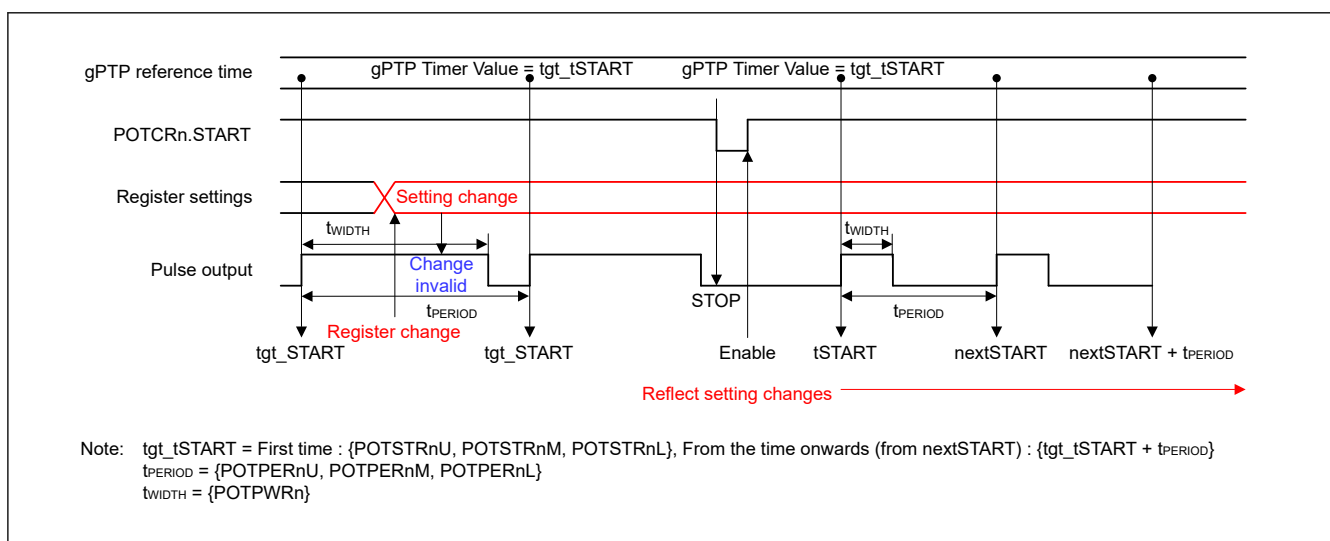


Figure 35.20 When the register setting is changed with the POTCRn.START bit set to 1

35.6.5 Interrupt Output

Interrupt output detects the rising edge of the pulse output and outputs an interrupt signal synchronized with PCLK.

There are four channels of interrupt output, and the interrupt signal corresponding to each pulse output channel is output. The interrupt signal outputs H with a width of 1 cycle of PCLK.

Figure 35.21 shows the timing diagram of interrupt output.

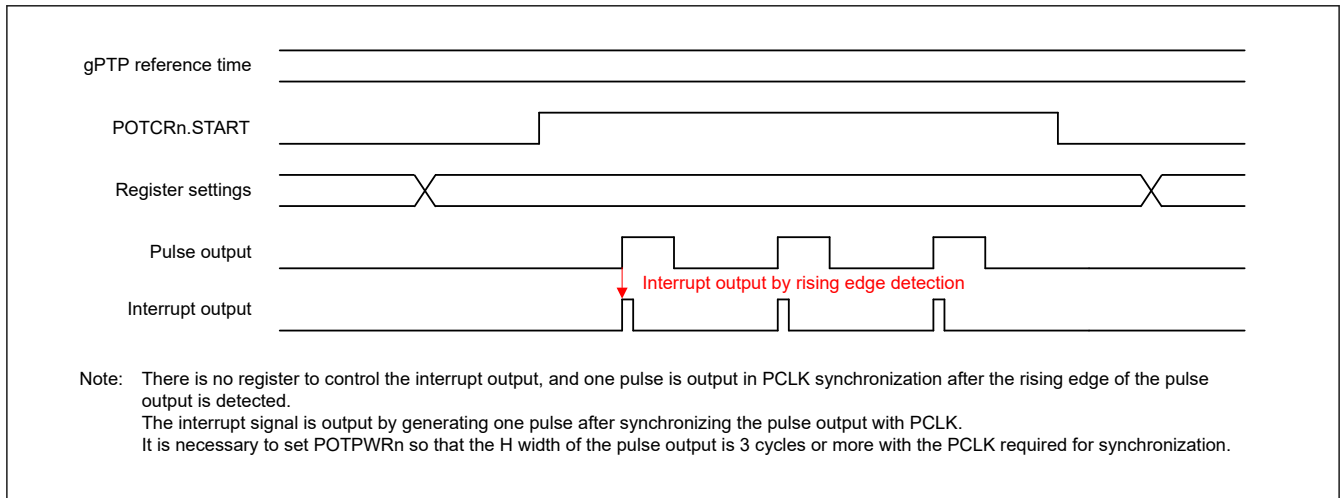


Figure 35.21 Detects the rising edge of the pulse output and outputs one pulse synchronized with PCLK

35.6.6 Precautions

35.6.6.1 Register Setting

The POTCFGR register switches the GPTP timer selection immediately when the setting is changed regardless of whether it is stopped or operating.

Be careful when changing setting during operation.

35.6.7 Restrictions (Including Known Problems)

35.6.7.1 Register Setting

The following registers should be set within the limit range of the set value:

- POTSTRnL, POTPERnL: Set the value in the range of 0 - 0x3B9AC9FF. 0x3B9ACA00 is 1 second, so if it is 1 second or longer, reflect and set it in the register in seconds.
- POTPWRn: For settings other than 0, set for a time of 3 or more cycles of the PCLK cycle. The time required to generate the interrupt output signal.